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Compute Project

Open Rack V3 Blind Mate Quick Connector Specification (BMQC)

Revision 1.0

Authors: Glenn Charest – Meta, John Fernandes - Meta, Nick Goenner – SafeWay,
Chris Lee – Wiwynn, Tim Marquis – Parker, Noman Mithani – Meta, Andy Holst –
Parker, Emil Pettersson – CEJN, Amit Shende – Danfoss

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1.2 Acknowledgements

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Parker - Steve Powell, Jeff Grau

2. Compliance with OCP Tenets

Please describe how this Specification complies to the following OCP tenets. Compliance is required for at least three of the four tenets. The ideals behind open sourcing stipulate that everyone benefits when we share and work together. Any open source project is designed to promote sharing of design elements with peers and to help them understand and adopt those contributions. There is no purpose in sharing if all parties aren't aligned with that philosophy. The IC will look beyond the contribution for evidence that the contributor is aligned with this philosophy. The contributor actions, past and present, are evidence of alignment and conviction to all the tenets.

2.1. Openness

This specification shares the key design interfaces and requirements to enable others outside of the group to use and adopt the valve designs for their application.

2.2. Efficiency

The sharing of this specification will allow others to leverage the components and save development time.

2.3. Impact

This specification and the development work performed by the suppliers in this group makes available to the community interoperable valves with the potential for multiple sources and leverage economies of scale.

2.4. Scale

The development of this specification with multiple suppliers that are interoperable will enable the solution to be deployable at scale.

2.5. Sustainability

This specification along with related specifications & guidelines for Orv3 Blind Mate will enable liquid cooling at scale will help reduce power consumption.

3. Version Table

Date	Version #	Author	Description
04 June 2024	1.0	See Authors	Initial Release of Specification

4. Scope

This document defines the technical specifications for the blind mate quick connectors (BMQC) used in Open Compute Project Open Rack Version 3. This is a specification developed with industry partners to standardize the form, fit, and function of the valves so that they can interoperate between the different suppliers.

5. Overview

One of the enhancements for Open Rack V3 (ORv3) over the previous rack is the support for liquid cooling. Blind mate liquid cooling reduces service times and can have layout advantages vs. manual connections. This specification was developed to create blind mate quick connectors that are supplier interoperable. The interfaces and design information for the IT gear, rack interfaces, and manifolds for ORv3 are outlined elsewhere in other specifications and design guides. The ORv3 solution has unique challenges due to the serviceability & scalability focus inherent in the frame design.

The blind mate valve design in this specification, referred to as BMQC (blind mate quick connector), features mechanical performance to adapt to these effects including enhanced radial misalignment allowance, angular offset allowance, gatherability, and stroke. The drivers of the valve misalignment and stroke (working range) requirements include IT gear to rack interfaces and gaps, frame manufacturing methods, valve mate forces, deflections from loads, coefficient of thermal expansion, and manufacturing tolerances of the various components and subassemblies.

The solution is self-contained with no add on components to achieve the float and stroke performance. Note that due to the floating nature of the solution, internal tubing/hose selection that connector to the plug and tubing management (internal to IT gear) are critical to the solution; more details are discussed later in this document.

This specification is part of a family of specifications and guidelines that define the requirements for ORv3 Blind Mate Liquid Cooling and include:

- ORV3 Blind Mate Quick Connector Specification (this document)
- Open Rack V3 Base Specification
- Meta Open Rack Frame V3 Specification
- Open Rack V3 Blind Mate Manifold Base Specification
- Open Rack V3 IT Gear Design Guide

The tolerance and deflection requirements for the IT gear, manifolds and frame are each defined in their respective specifications or guidelines. They include:

- Meta Open Rack Frame V3 Specification
- Open Rack V3 Blind Mate Manifold Base Specification
- Open Rack V3 IT Gear Design Guide

Note that collateral is included with this specification and includes the following:

- Reference 3D design step files of the valves (see Section 8.0 for details)
- Test Fixture guideline (see Section 12 for more details)
- Self-Centering check tool cad (see Section 9.2.2 for details)
- Plug Installation tool reference design (see Section 7.5.2 for details)
- PTFE tubing design drawing of tubing used for testing.

6. Rack Compatibility

The valve design is designed to work with the Meta Version of the ORv3 Rack, OCP ORv3 Blind Mate Manifolds, and gear designed per the Blind Mate IT Gear Design Guide. However, end users may find that it can be used in other applications.

7. Design Basics

This section provides the design fundamentals of the valve designs for the plug and socket, mating stages, and interface requirements. Additional detailed design requirements are outlined later in the specification.

The blind mate valve (BMQC) designs outlined in this specification consist of a (1) socket valve and a (2) plug valve. In this application, the socket is to be mounted to a manifold and the plug is mounted to an IT gear chassis. The socket threads into the manifold and features ISO threads and O-ring. The plug installs to the chassis through a hole in the panel at the rear of the chassis from the inside and is fixed in place with a panel nut.

The valve pair can accommodate radial misalignment as well as combination of radial and angular misalignment within defined zones. Due to the relationship between radial and offset and angles (and the resultant projected differences), combined and angular and radial offsets are provided in an allowable offset zone and mate distance; see Section 9.1 for details. The socket valve features a cone to gather and align the pair as they connect.

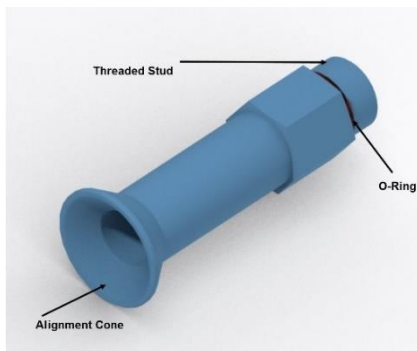


Figure 7.0.1 Socket

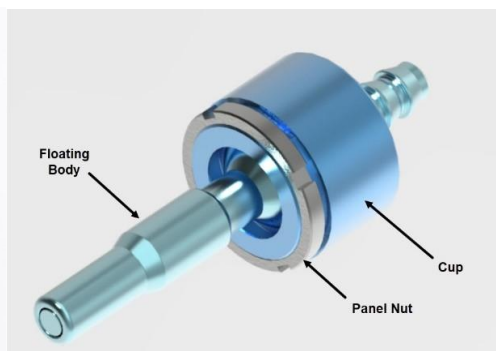


Figure 7.0.2 Plug

The plug features three main components: (1) the cup (2) the floating body, and (3) the panel nut. The plug floating body is designed to adjust for radial and angular misalignment as the 2 valves align and connect. The plug assembly incorporates spring(s) and centering features to help the plug floating body self-center when no external load is applied. In some solutions, there are rear centering features that are proud of the rear of the cup; more details on this area will be shared later in this specification. The centering spring(s) push the floating body out towards the socket; the details on how this capability is achieved are not defined in the specification and left

to the suppliers to develop. There are self-centering requirements with tubing attached and maximum spring force for the centering spring are both outlined later in this specification. The rear side of the plug floating body features a barb interface to the internal tubing. The plug design is sensitive to tubing stiffness so careful consideration of tubing selection and routing in chassis are required; see Section 10 for more details.

During mating, the plug and socket make contact and the mate forces push the floating body of the plug away from the socket eventually clearing the internal plug centering features until it hits an internal rear stop in the plug cup (this distance is 4.5mm). Full movement of the plug floating body is guaranteed once the floating body hits the internal stop at the back of the plug cup (note that full movement may be reached prior, but this is left to the suppliers to determine). Smooth transition from centered to full floating is critical for smooth operation. Later in this document, preliminary force requirements are provided in various conditions.

7.1 Basic Geometry

7.1.1. Socket

Basic geometry for the socket half of the BMQC is provided in Fig 7.2.1 below. Datum A on the socket is the surface that sits flush with the manifold spot face.

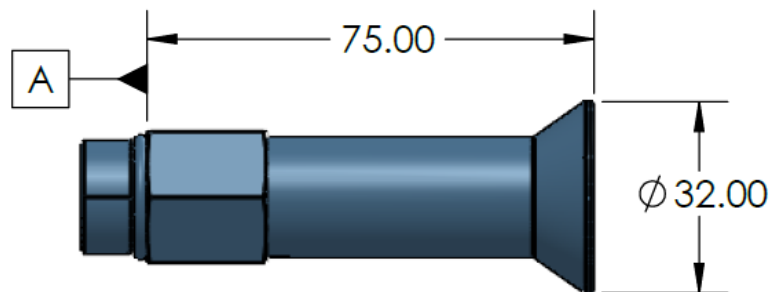


Fig 7.1.1 Socket half with basic dimensions and features

7.1.2 Plug

The view in Fig 7.2.1.2. shows the plug in the unloaded condition. Note that the floating body of the plug moves 4.5mm axially toward the inner stop in the cup during connection. Datum A on the plug is the cup surface that is flush with the inside of the sheet metal mounting bracket on the IT Gear. Table 7.2 provides dimensions for the protrusion length from Datum A for the loaded and unloaded states. In some solutions, there are rear centering features that are proud of the rear of the cup.

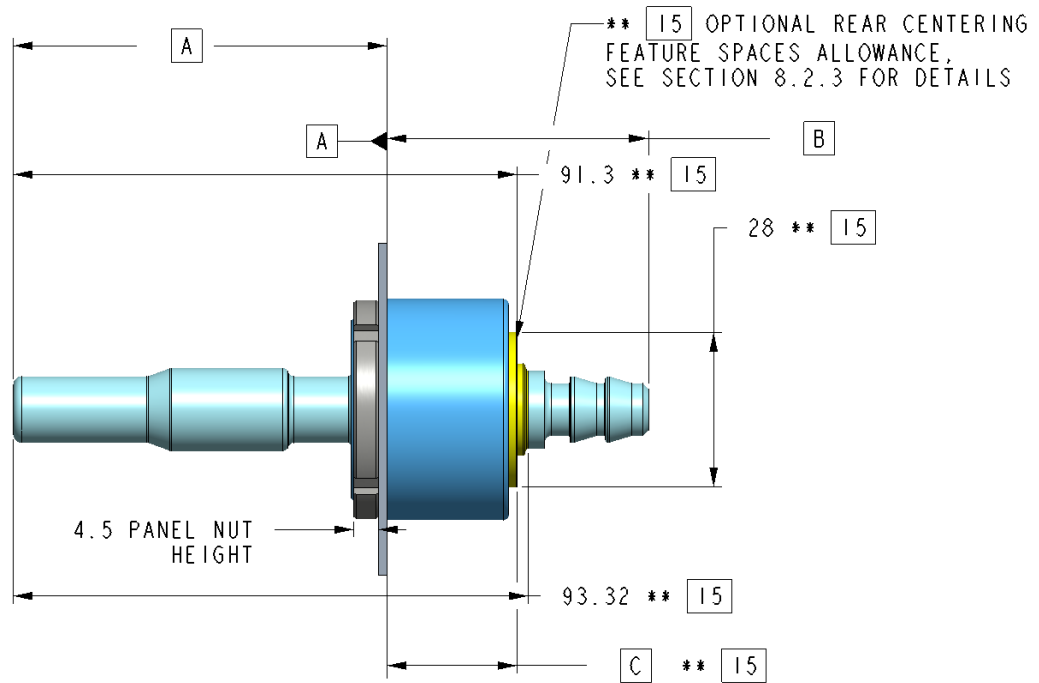


Fig 7.1.2 Plug with key dimensions and features in Unloaded State
 **See section 8.2.3 for details for this allowance area

Table 7.1 Plug Tip Length Protrusion

State	Dimension A	Dimension B	Dimension C
Unloaded	67.7	47.4	23.6
Loaded	63.2	51.9	28.1

7.1.3 Mated Assembly

In the figure below, a mated valve pair are shown in a side view with various reference dimensions. The mate distance dimension below is from the socket valve Datum A (coplanar with the manifold spot face) to the inner chassis plug mounting surface (plug valve datum A).

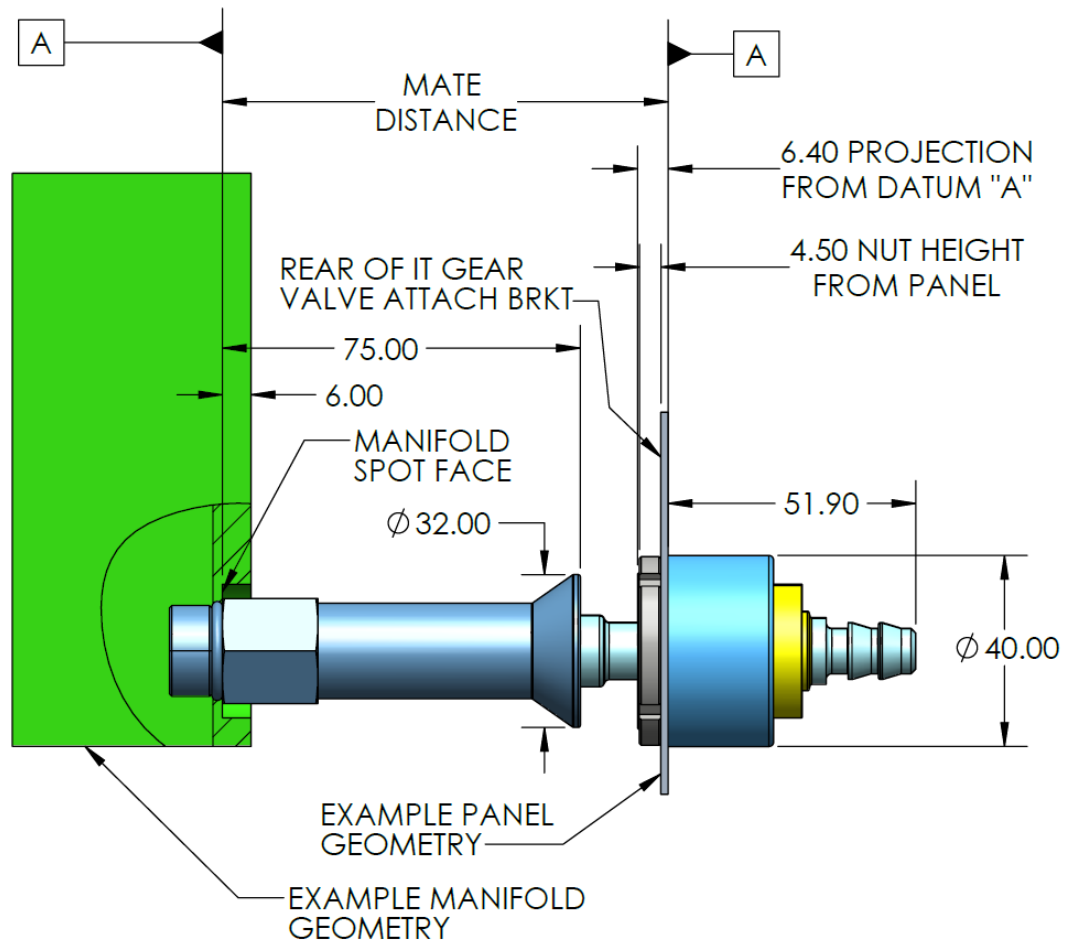


Figure 7.1.3 Mated assembly depicted with example IT Gear panel and manifold.

7.2 Mating Stages

This section describes the various mating stages of the valves. These stages are based on key mate distances where important interactions occur between valves as they mate. Parameters such as tolerances, deflections from loads, and misalignment will affect the exact mate distances at each stage; the values provided are for reference to help one understand how the valves interact at various mate distances. Also, note that depending on friction and loading, the plug floating body may push into the cup at earlier stages than shown. Note that tubing connected at the barb end and has been removed for simplicity in all views.

7.2.1 Stage (1) - “Initial Contact”:

This is the stage where the plug and socket make initial contact. The value for contacts can vary depending on misalignment and plug skew. In this example, the socket and plug are offset to the maximum allowable offset and the floating body of the plug is at the maximum allowable skew offset with no load.

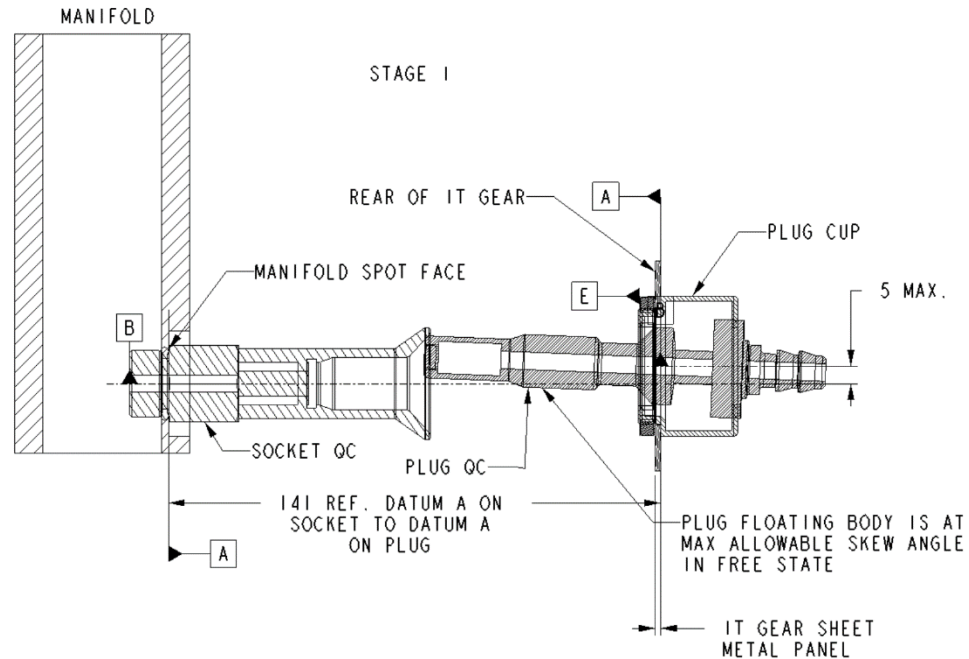


Figure 7.2.1 Stage (1)

7.2.2 Stage (2) - “Gathered”:

The plug is now gathered and centered into the socket.

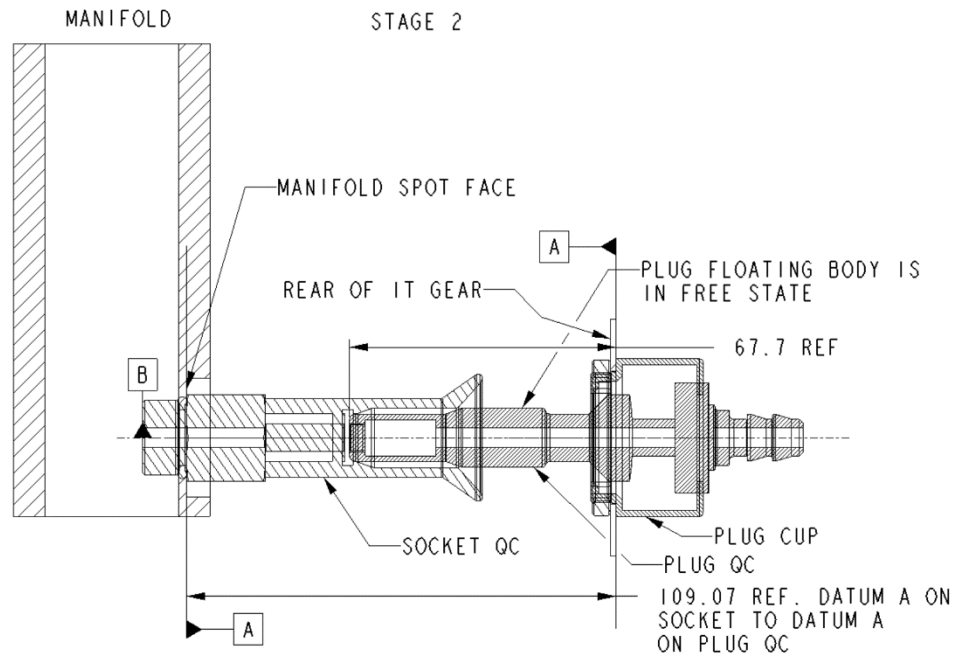


Figure 7.3.2 Stage (2)

7.2.3 Stage (3) - “Flush”:

The plug and socket valve faces are now flush.

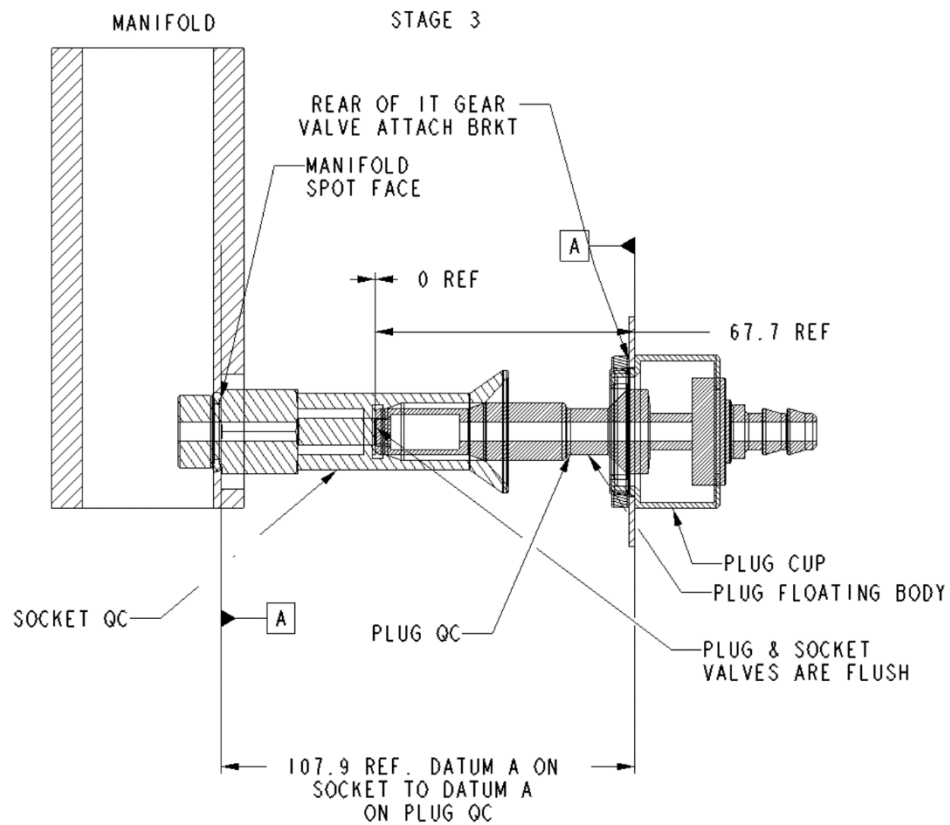


Figure 7.2.3 Stage (3)

7.2.4 Stage (4) - "Flush & Plug Pushed In":

The plug floating portion is pushed into the IT gear chassis until it hits the plug internal stop. Note that depending on friction and seal forces, the plug floating part could reach the stop earlier in mate process.

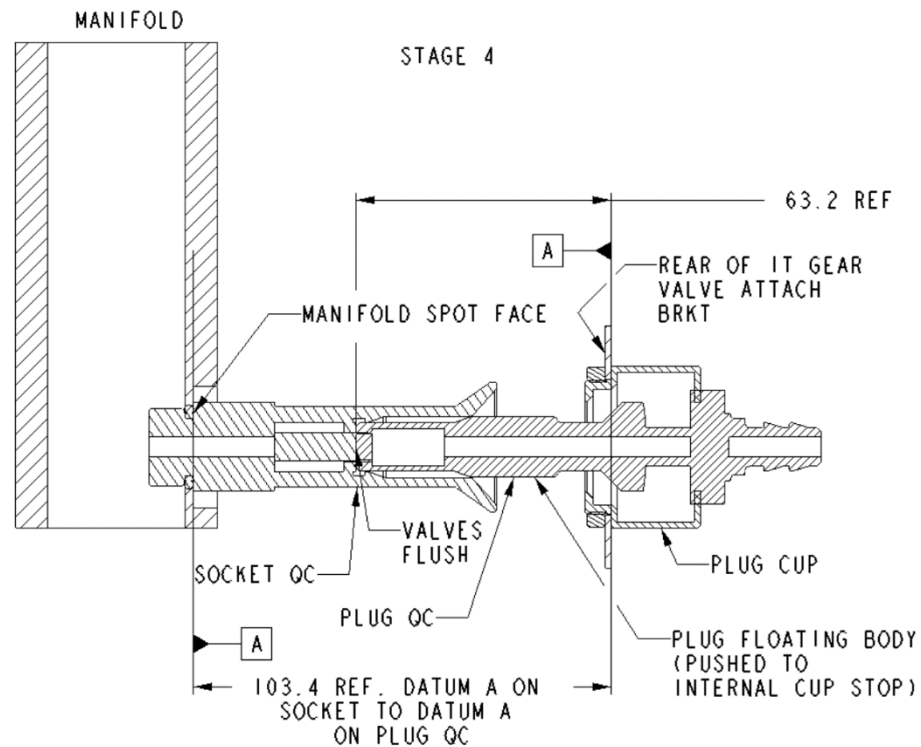


Figure 7.2.4 Stage (4)

7.2.5 Stage (5) - “Maximum Mate Distance”:

Between stage 4 & 5, the valves move closer together and the valves start to open. At stage 5, the valves are fully open. This is the largest mate distance that the valves will perform at full flow between the 2 valves. Care shall be taken by the developer to ensure that the design solution prevents the valves from mating any distance beyond the specified value. The valves will start to close, and performance will be affected beyond this value. Note there is no specific flow requirement for the design at mate distances larger than this value. The value is also listed in table 9.3.

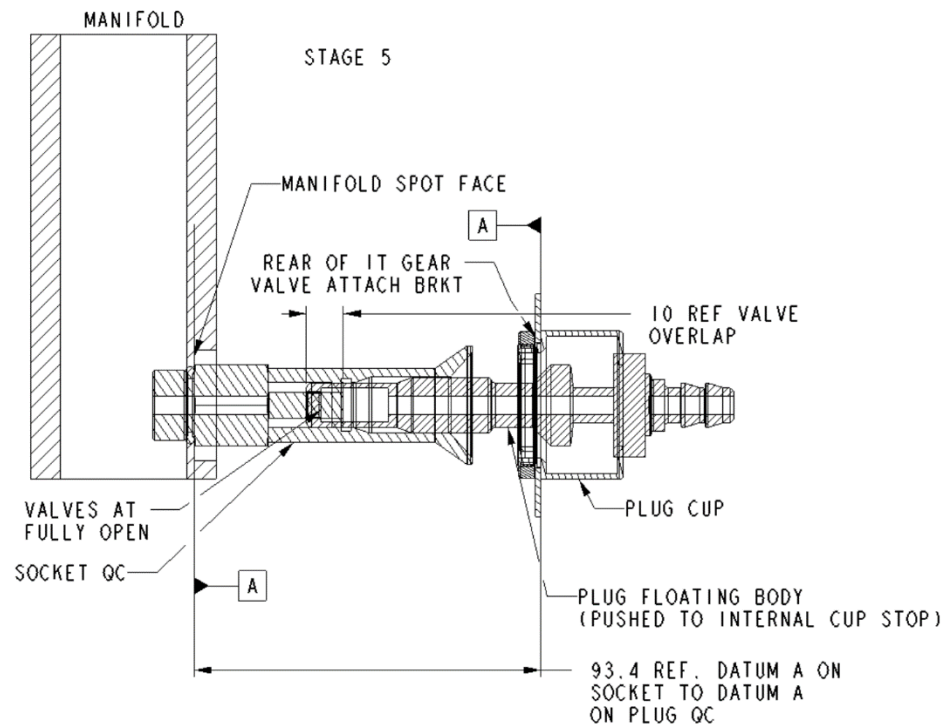


Figure 7.2.5 Stage (5)

7.2.6 Stage (6) - “Nominal Mate Distance”:

This is the nominal mate distance between the valves if the IT gear injector levers are flush against the Datum A catch edge features in the ORv3 frame. This figure assumes the various components (frame & interfaces, IT gear, and manifolds are designed per their respective specifications or guidelines).

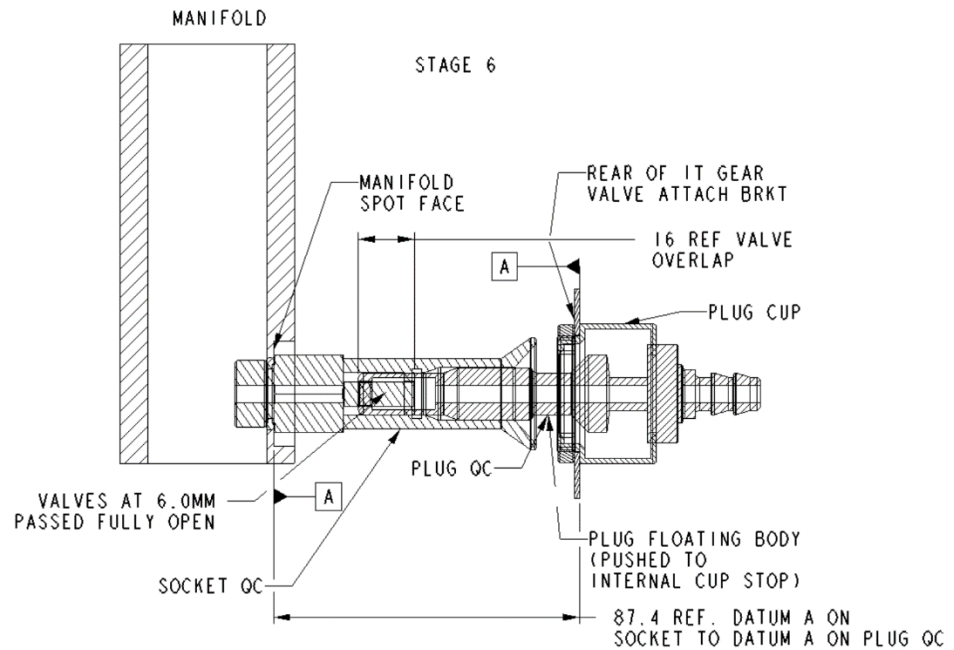


Figure 7.2.6 Stage (6)

7.2.7 Stage (7) - “Minimum Mate Distance”:

This is the minimum mate distance between the 2 valves. Care shall be taken by the developer to ensure that the design solution prevents the valves from mating any distance less than specified to avoid damage to the valves. The value is also listed in table 9.3.

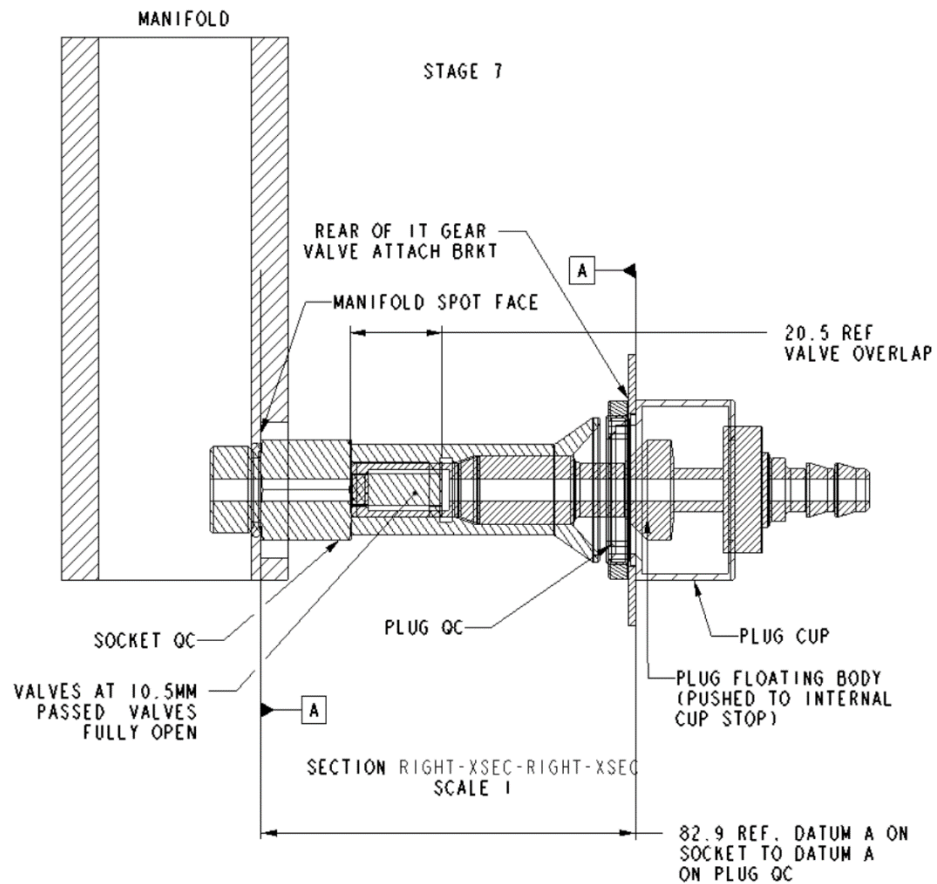


Figure 7.2.7 Stage (7) - Minimum Mate Distance

7.3 Mating Interfaces

7.3.1 Plug (IT Gear) Interface

The BMQC plug mounts to a panel and the panel information is outlined below. Details on the location of the valves with respect to the chassis and the recommended keep out areas are provided in the Open Rack V3 IT Gear Design Guide. The recommended panel material properties: minimum Yield Stress 248MPa and minimum hardness 56 HRB. The developer shall ensure the panel meets the test listed in Test 11.1-13.

MOUNTING PANEL & HOLE DETAILS

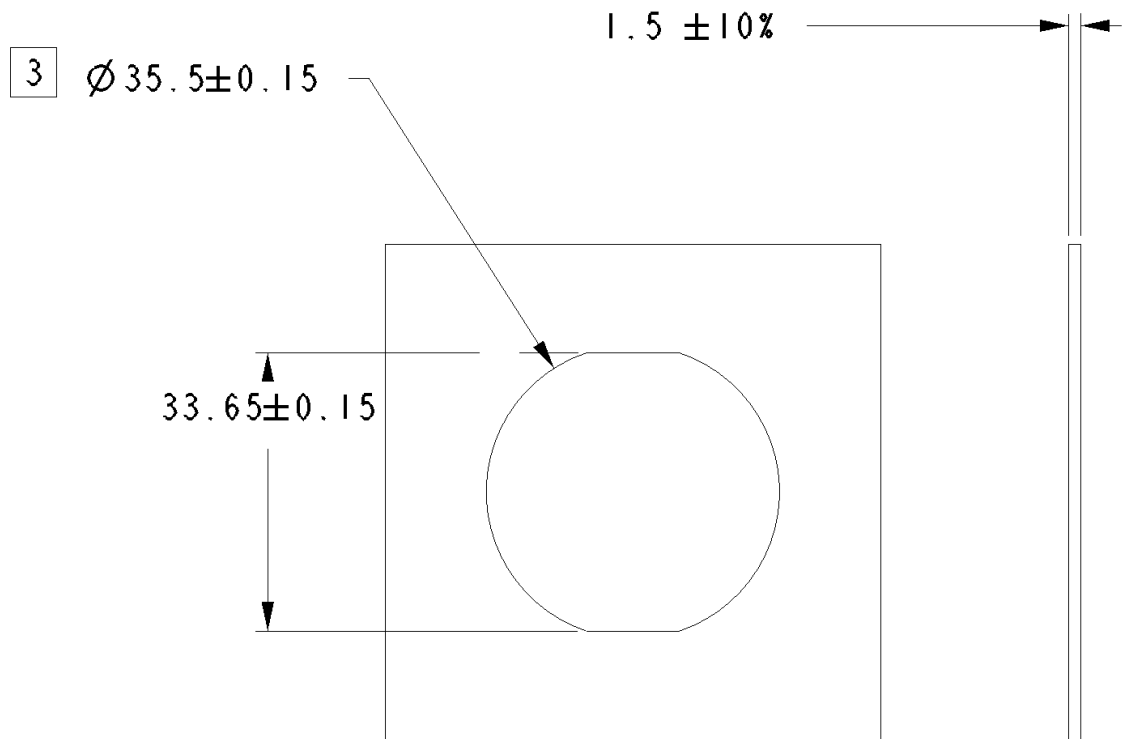


Figure 7.4.1 Plug Valve Panel Requirements

7.4.2 Socket Interface

The socket valve has a threaded stud end that is installed into a manifold. The stud & O-ring details for the socket are summarized below. Manifold Interfaces are listed in the manifold section. Socket Interface Features:

- STUD END ISO11926-3:1995(E) - 3/4-16 NON-ADJUSTABLE
- O-RING PER ISO11926-3:1995(E) - 3/4-16

7.4.3 Plug Chassis Space Considerations.

As mentioned in other sections, during development of the valves, some suppliers within the group added features in their designs to the rear area of the floating body between the barb area and the area that sits in the plug cup to help with self-centering. These features protrude past the main cup depth wise and

during use can also exceed the diameter of the plug cup during operation at radial and combined offset conditions due to tolerances and dynamic effects during alignment to mate. The images below show these centering feature space allowances in yellow. Developers shall make sure there is no interference with these features in their application. Below are some reference views and comments to consider when mounting a plug in a chassis. More details on hose routing and chassis guidance are provided in Section 10 and in the Open Rack V3 IT Gear Design Guide.

Below, in Figures 7.4.3.1 & 7.4.3.2 are views of plug with the floating body offset at two maximum allowable conditions outlined in Section 9. The views show the location of the rear centering features volume (in yellow) as these offsets cup. The dimensions annotated with “**” show the zone that these features occupy at the offset conditions. The dimensions are shown for reference and when determining the appropriate clearances, developers should account for tolerances, deflections, CTE, and dynamic effects of plug motion during alignment. Actual deflections will be subject to the developer’s own design and analysis.

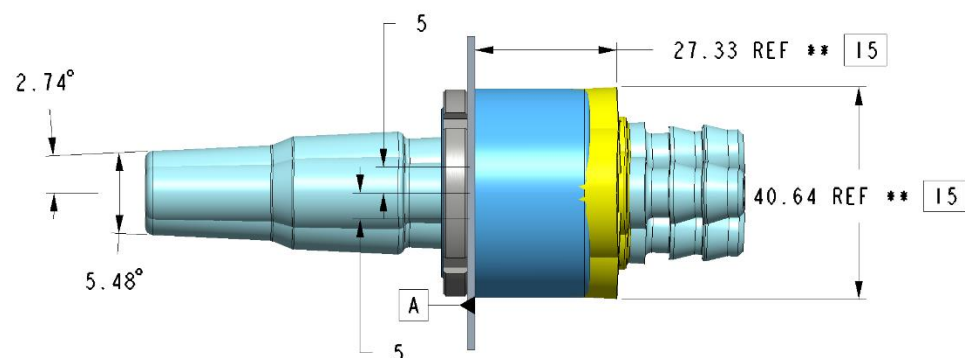
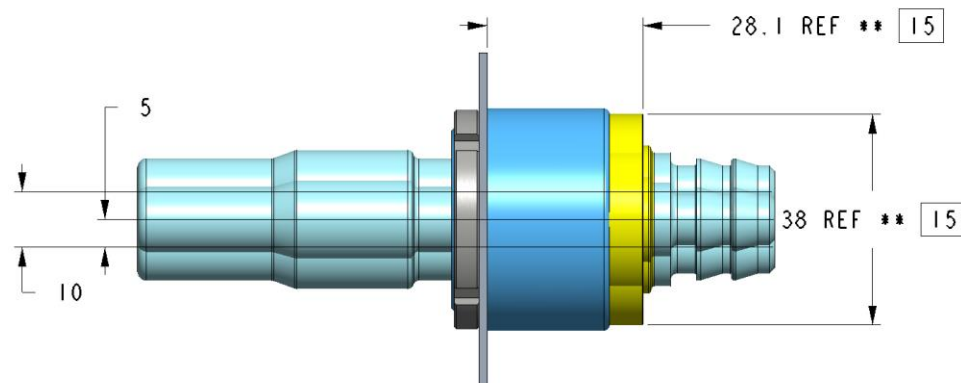


Figure 7.4.3.1 – Combined 5 mm & 2.74° angle offset

Figures 7.4.3.3 below illustrates the nominal location of the optional rear centering features (in yellow) at the maximum allowable offset (at 5mm & 2.74° angle offset) looking from the rear. For ease of illustration, the figure below shows the largest nominal anticipated deflection of the centering features (40.64mm) occurs at approximately 45° between the pure horizontal and vertical planes; this largest offset is the resultant of the horizontal (X) and vertical tolerances (Z). See Fig 7.4.3.4 for additional context. Actual deflections will be subject to the developer's own design and analysis.

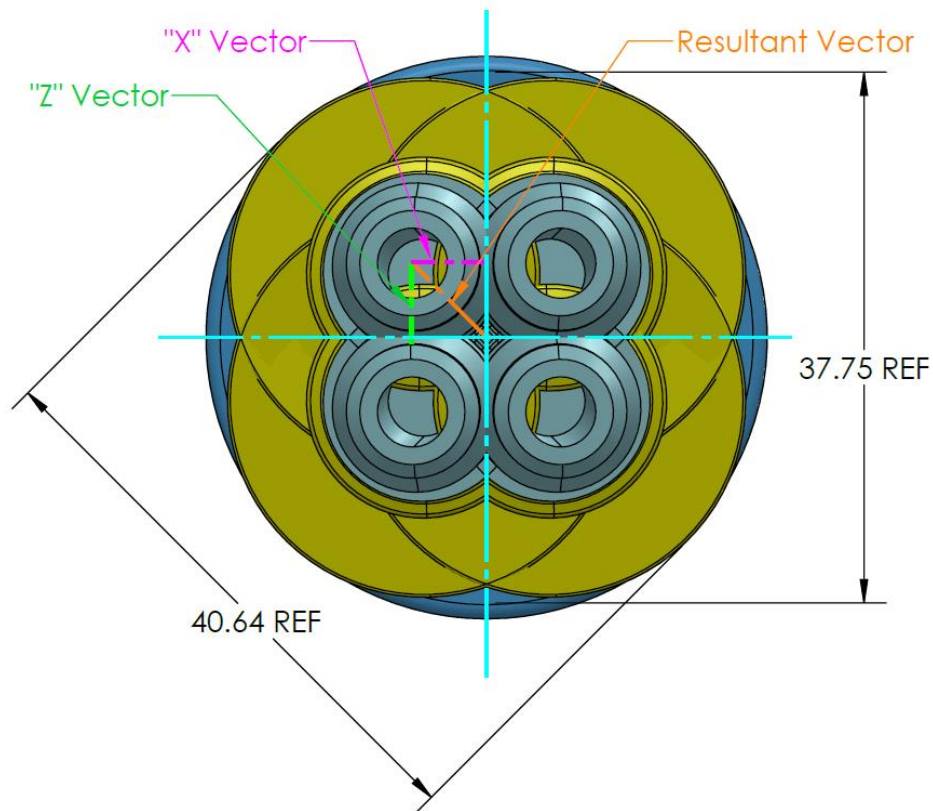


Fig 7.4.3.3 - Rear View of Plug Assy with several floating bodies at Various Potential locations at the Combined 5mm & 2.74° angle offset

Figure 7.4.3.4 below is a simplified view of the same conditions (5mm & 2.74° angle offset) showing the tolerance vectors and the resultant total offset. A simplified outer zone for the nominal location of the optional centering features is also shown.

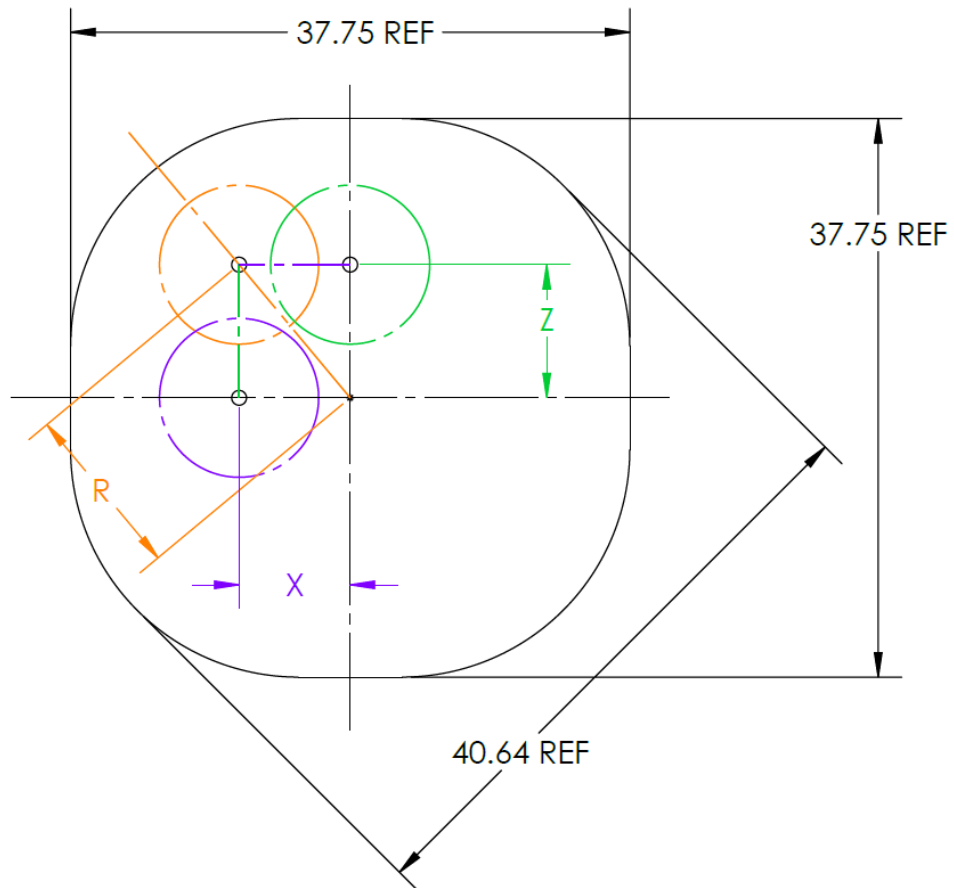


Fig 7.4.3.4 Simplified Rear View of Plug Assy Diagram at the Combined 5mm & 2.74°

7.5 Installation

7.5.1 Installation Torque

Torque specifications are listed below for tightening torque for each valve.

Table 7.5.1: Torque Recommendations

Plug Install with Panel Nut	25N-m +/- 5N
Socket Install into manifold	40N-m +/- 5N

7.5.2 Plug Installation Tool

To tighten the panel nut of the plug valve, a special tool is needed that is designed to clear the plug and interface with features on the panel nut. A

reference design is provided in CAD and works with a “1/2” inch ratchet. 3D CAD (STEP format) of the tool is provided for reference with this document.



Figure 7.5.2 Various Views of the Plug Installation Tool

7.5.4 Socket Installation

The socket is threaded into the manifolds and torqued to a maximum torque listed in Table 7.5.1 (threaded area is shown below in green). The base of the socket features hex nut features (surfaces in red in figure below) requiring an open torque wrench for installation.

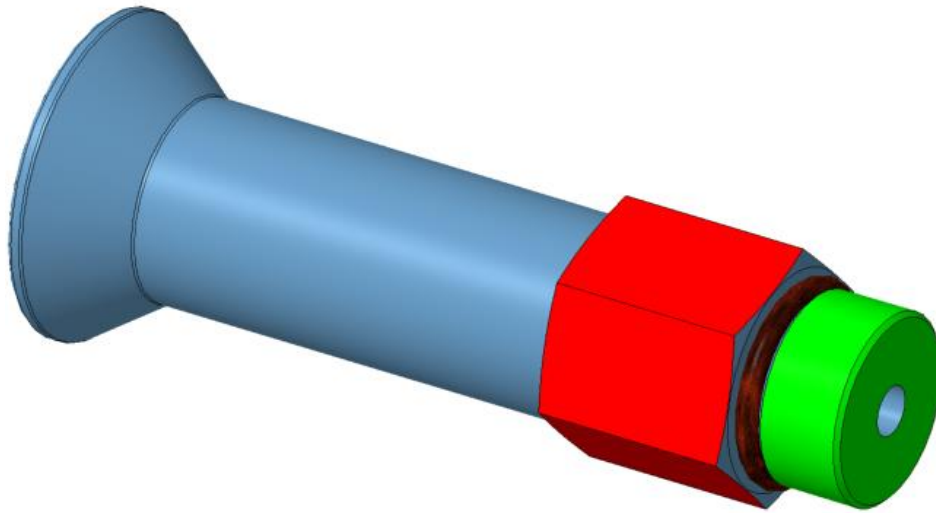


Figure 7.5.4

7.5.3 Plug Installation

The plug assembly installs into the panel from the inside of the chassis and is retained by a threaded nut that is installed and tightened with a special socket tool. The installer inserts the plug through the panel hole until the cup Datum A touches the inside panel surface, then holds the valve main body (cup) firmly against the panel, then rotating the valve body (cup) until the valve body flats align with the flats on the panel hole) allowing the plug to sit flush with the inside wall of the chassis, and then tightens the nut onto the plug body (figure 7.4.1). The panel flats in the hole interface with flats on the plug body to prevent the valve from spinning when tightening the nut to the panel. The panel nut maximum tightening torque is listed in table 7.5.1. Installation of the tubing on the back of the plug is covered in Section 10.5.

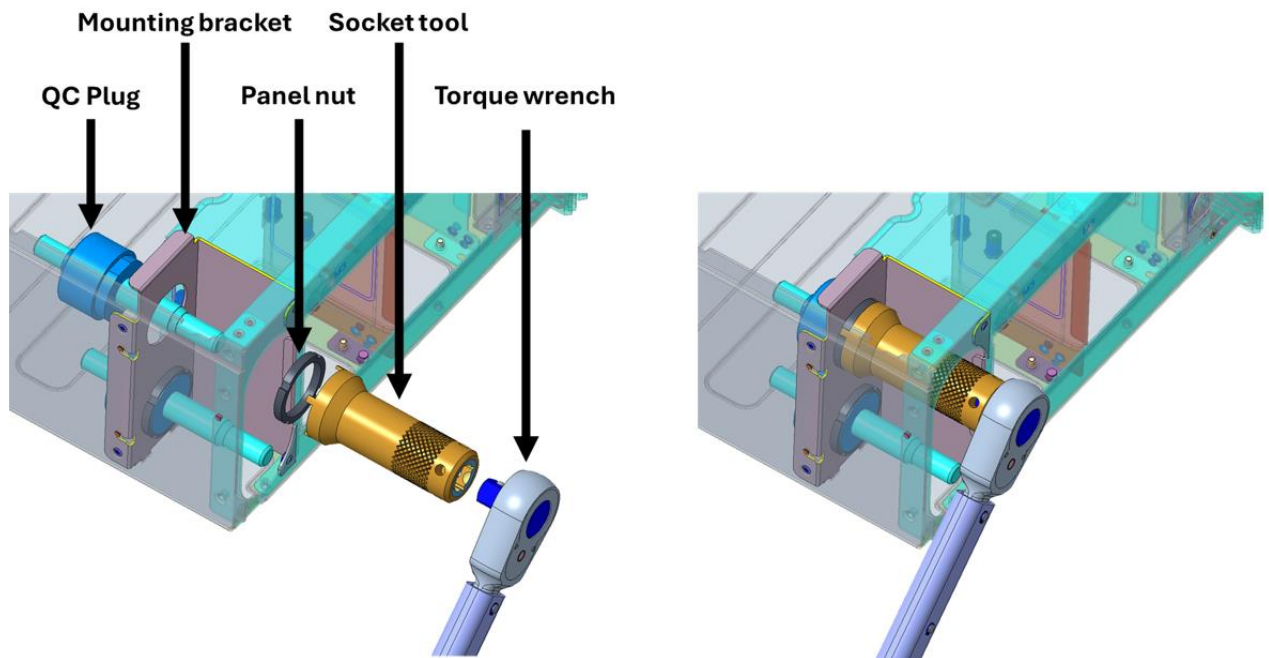


Figure 7.5.3 Installed plug on the mounting bracket by locknut, socket tool and torque wrench

8. Mechanical Design Specifications

This section provides the detailed design requirements for the various components and assemblies that make up each valve (socket and plug). The designs are not fully defined, rather key interfaces and features are defined. The valve suppliers in the group wanted the specification to allow freedom to develop their own unique solutions as well as protect their IP in terms of internal valve design features. Figure 3 below has the notes that apply to all detailed views in this section. 3D CAD step files for the plug and socket are provided in the collateral with this specification. Note that the CAD provided is for reference and does not have the design details of each supplier's unique features. See figures below for all critical to function details and requirements.

Open Compute Project • Orv3 Blind Mate Quick Connector Specification (BMQC)

1. INTERPRET DIMENSION AND TOLERANCES PER ASME Y14.5-M 2009.
2. ALL DIMENSIONS ARE IN MM. 3D CAD FILE ALSO PROVIDED FOR REFERENCE.
3. DEFINED DIMENSIONS AND TOLERANCES, INCLUDING GD&T CONTROLS, TO MEET $C_{pk} \geq 1.33$.
4. ALL COMPONENTS SHALL BE ROHS AND UL COMPLIANT. ALL SEALS TO BE $< 0.4g$ (EACH SEAL).
5. ALL PARTS AND COMPONENTS TO BE FREE OF BURRS AND SHARP EDGES.
6. ALL EDGES OR INTERFACES SURFACES TO BE RADIUS TO REMOVE BURRS AND SHARP EDGES TO PREVENT SCORING AND DAMAGE TO SEALS WHEN MATING. SIZE RADIUS PER SUPPLIER PREFERENCE.
7. SURFACE FINISH CALLOUTS SHALL BE INTERPRETED PER ASME Y14.36M and ASME B46.1 STANDARDS. THE SPECIFIED REQUIREMENT REPRESENTS TO MAXIMUM ROUGHNESS AVERAGE (R_a) ANYWHERE ALONG THE INDICATED SURFACE.
8. STUD & O-RING DIMENSIONS SHOWN FOR REFERENCE, CONSULT ISO 11926-3:01995(E) SIZE 3/4-16 FOR APPROPRIATE REQUIREMENTS. STUD IS NON-ADJUSTABLE.
9. DETAILS INSIDE THE SOCKET & PLUG INTERNALS NOT SPECIFIED AND ARE TO BE DESIGNED TO SUPPLIER'S PREFERENCE WHILE MEETING THE VARIOUS REQUIREMENTS.
10. DETAILS FOR FEATURES INSIDE OF PLUG CUP AND THE PORTION OF THE PLUG FLOATING PART THAT SITS IN THE CUP WHEN ASSEMBLED ARE MODELED FOR VISUAL REFERENCE. THESE AREAS ARE TO BE DESIGNED TO SUPPLIER'S PREFERENCE WHILE MEETING THE VARIOUS REQUIREMENTS.
11. RETENTION JAM NUT TO BE INCLUDED WITH ASSEMBLY.
12. PARTS TO BE SUPPLIED WITH DUST CAPS ON BOTH ENDS TO PROTECT THE PARTS. KEY DIMENSIONS ARE PROVIDED FOR AREAS WHERE THE CAP SIZE AFFECTS INSTALLATION OR SERVICING. COVERS ON REAR OF SOCKET AND REAR OF PLUG TO FULLY COVER THREADS OR BARB FEATURES TO PREVENT DAMAGE.
13. DEFINED ASSEMBLY DIMENSIONS DO NOT REQUIRE MEASUREMENT AFTER ASSEMBLY BUT COMPONENT FEATURES THAT AFFECT THIS DIMENSION SHALL BE CONSIDERED KEY CHARACTERISTICS PER NOTE 3 TO ENSURE ASSEMBLY LEVEL TOLERANCE IS HELD.
14. MARK PARTS IN AREAS SHOWN. MARK PARTS IN AREAS SHOWN. OPTIONAL AREA DESIGNATED ON FLOATING BODY MUST NOT CONTACT SOCKET SURFACES UNDER ANY MATE CONDITION. MARK PART WITH: (1) SUPPLIER NAME, (2) PARTNO, & (3) BUILD DATE. SUPPLIER NAME CAN BE ABBREVIATED TO TWO LETTERS THAT INCLUDE THE FIRST AND LAST LETTER OF SUPPLIER NAME: SUPPLIERNAME-PARTNO-DATECODE
15. THIS AREA IS AN OPTION SPACE ALLOWANCE FOR CENTERING FEATURES THAT IS NOT PREFERRED ON FUTURE DESIGNS. SEE SECTION FOR DETAILS 8.2.3
16. FIRST ARTICLE INSPECTION REPORT TO BE PERFORMED ON ALL DIMENSIONS CALLED OUT IN THE SPECIFICATION FOR THE VALVE COMPONENTS AND ASSEMBLIES. ON A MINIMUM OF 3 SAMPLES. REPORT TO INCLUDE FOR EACH DIMENSION:
 - A. DESCRIPTION OF THE DIMENSION
 - B. NOMINAL, MINIMUM, MAXIMUM VALUES PER THE SPECIFICATION
 - C. MEASURED VALUE
 - D. UPPER/LOWER DEVIATION (%)
 - E. MEASUREMENT DEVICE
 - F. SAMPLE SIZE OR MEASUREMENT FREQUENCY (AS APPLICABLE)
 - G. MEETS SPECIFICATION VALUE - YES/NO
17. FOR PROCESS CAPABILITY STUDY, TO BE PERFORMED FOR ALL DIMENSIONS DESIGNATED PER NOTE 3. MEASURE MINIMUM 32 SAMPLES. INCLUDE THE SAME DETAILS AS NOTE 16 & ALSO INCLUDE FOR EACH DIMENSION:
 - H. CALCULATED C_{pk}
 - I. MEETS $C_{pk} 1.33$ - YES/NO

Figure 3 Drawing Notes

8.1 Socket Design Details

Below, are the various drawing views for the socket valve design with and without dust caps.

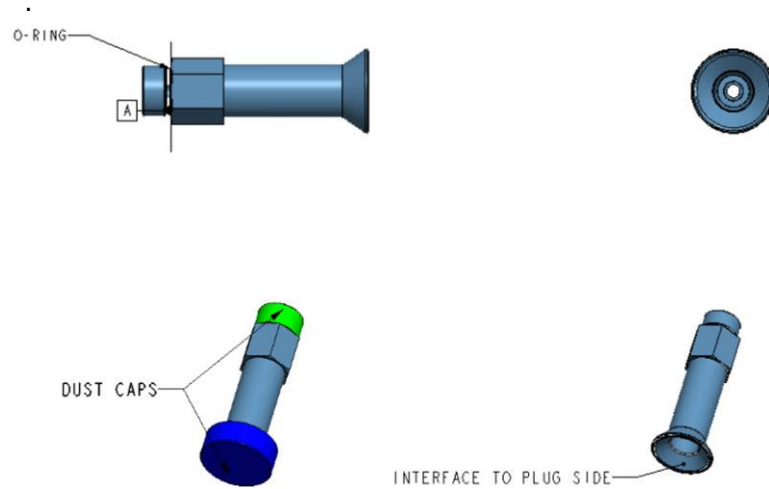


Figure 8.1.1 Socket Views

8.1.1 Dust Caps

Socket valves are to ship with dust caps. Below are the detailed size requirements.

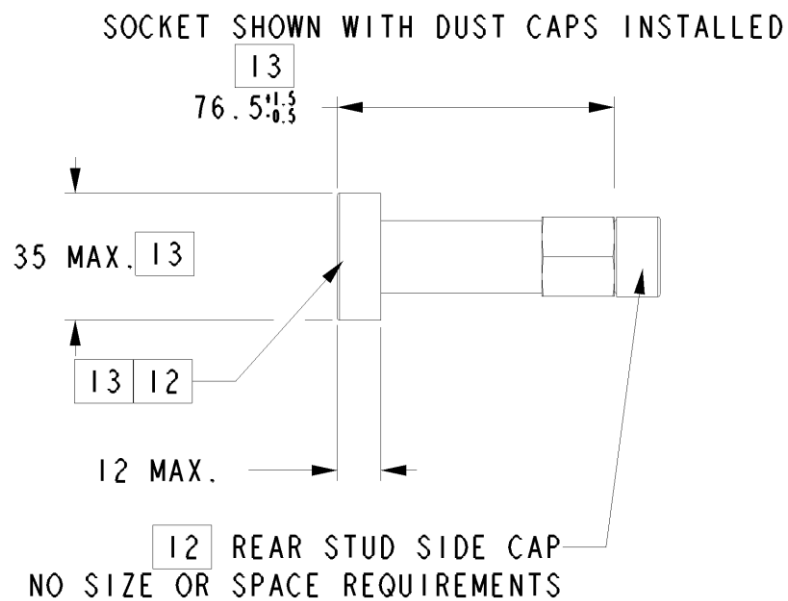


Figure 8.1.2 Socket Design Dust Cap Detail (Side View)

8.1.2 Detailed Socket Design

Below are the detailed mechanical design requirements for the socket valve.

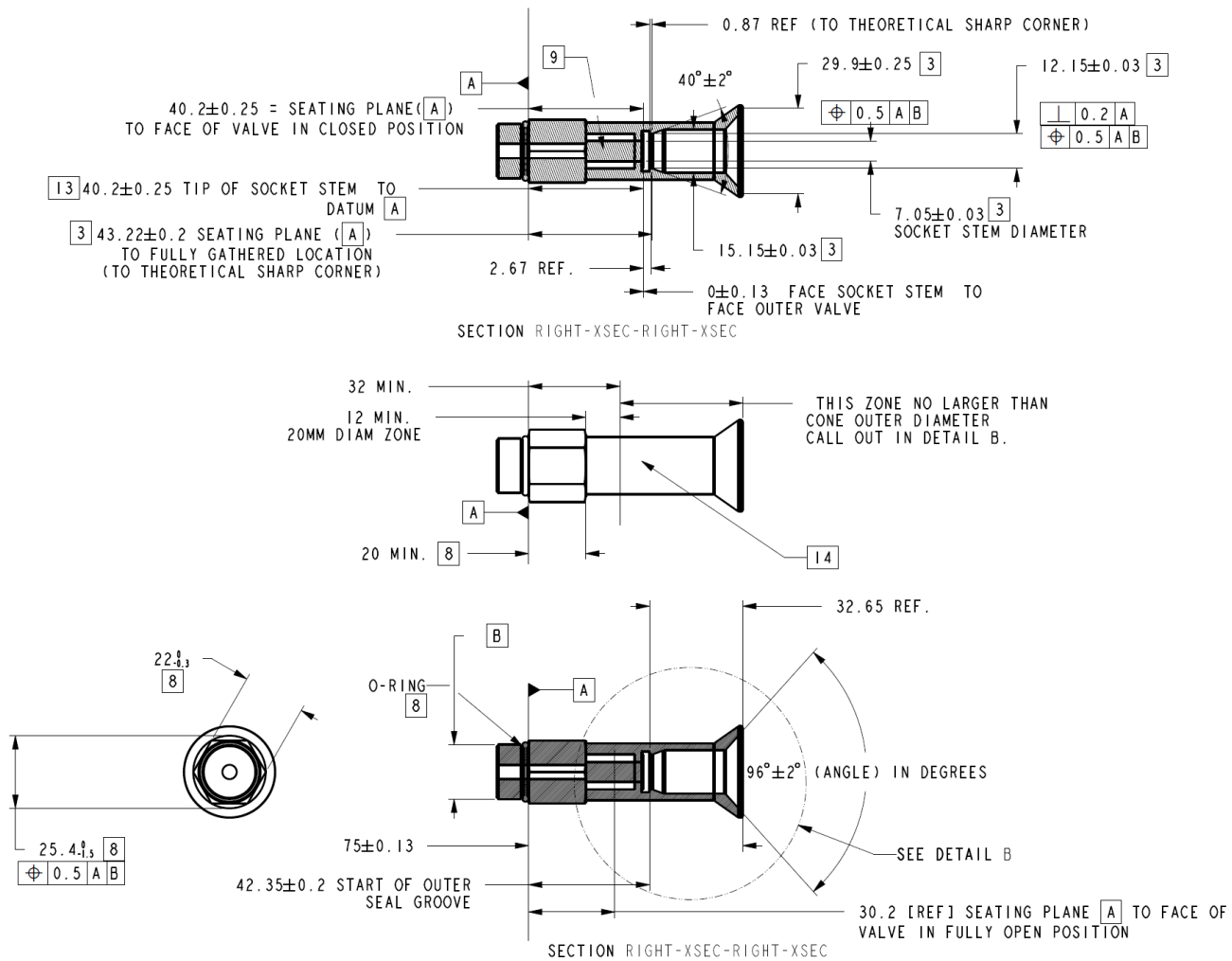


Figure 8.1.3

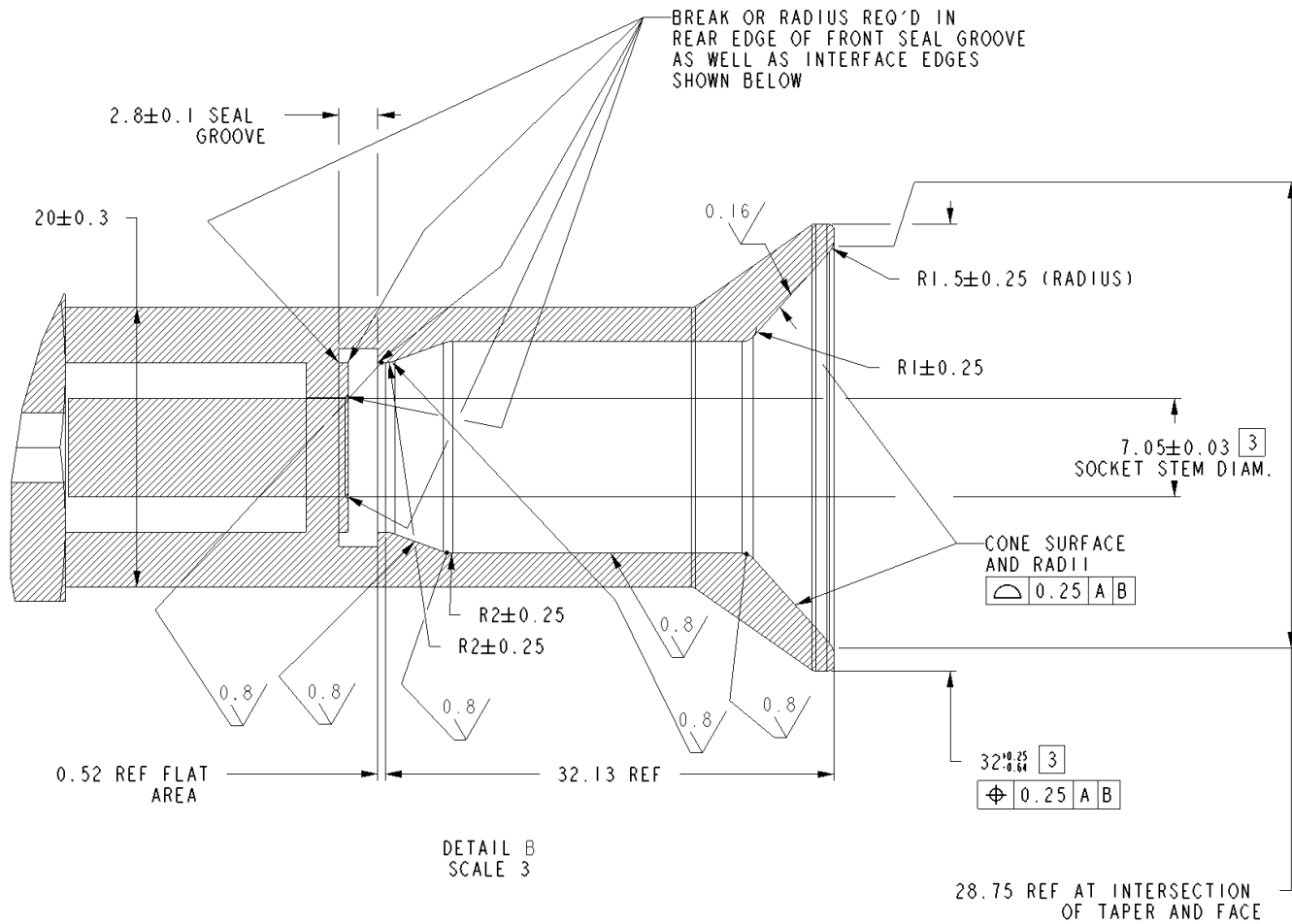


Figure 8.1.4 Socket Detail

8.1.3 Socket Weight

Table 8.1.3 Socket Weight

Maximum Socket Weight (excluding dust caps)	193g
---	------

8.2 Plug Design Details

8.2.1 Plug Assembly

Below, are detailed mechanical design requirements for the plug valve design. Requirements for both the unloaded state and the loaded state are outlined later in Section 9.2.

During development, some suppliers within the group added features in their designs to the rear area of the floating body between the barb area and the area that sits in the plug cup to help with self-centering. These features protrude past the main cup depth wise and during use can also exceed the diameter of the plug cup during operation at radial and combined offset conditions due to tolerances and dynamic effects during alignment to mate. These features move with the floating body as it moves. Developers shall make sure there is no interference with these features in their application; the recommendation on future designs to avoid any features that exceed the cup diameter to minimize clearance space requirements around the cup area. Dimensions in any views pertaining to this spatial allowance will be annotated below with “**”.

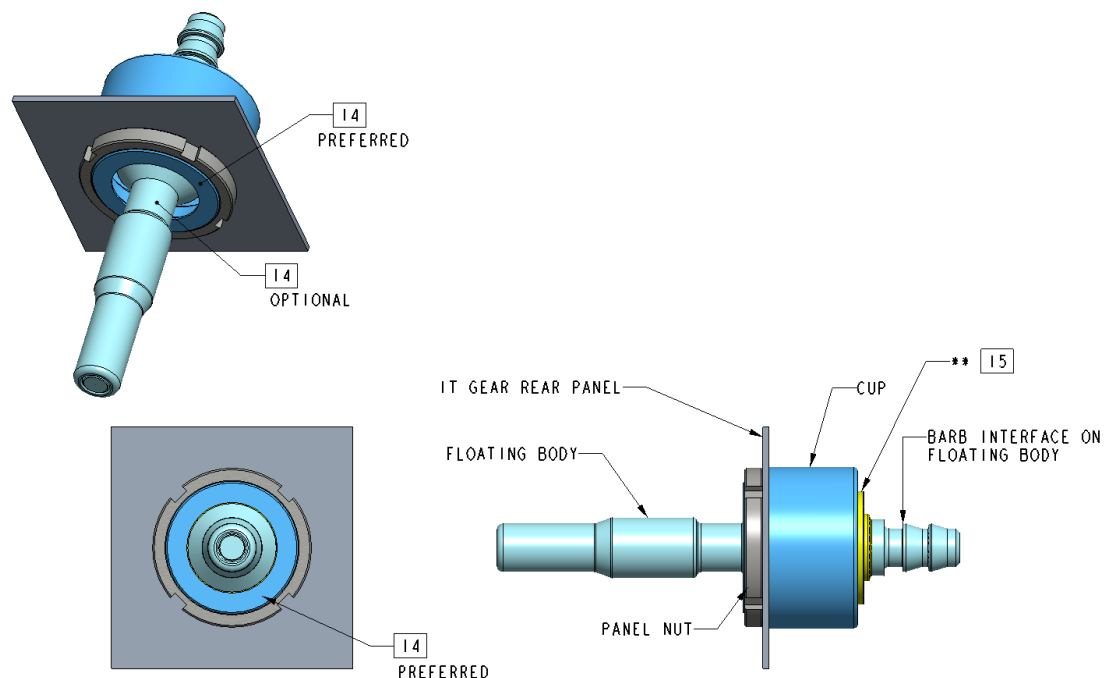


Figure 8.2.1 Plug Assembly in Free state

8.2.2 Plug Dust Caps

Plug valves are to ship with dust caps. Below are the detailed size requirements.

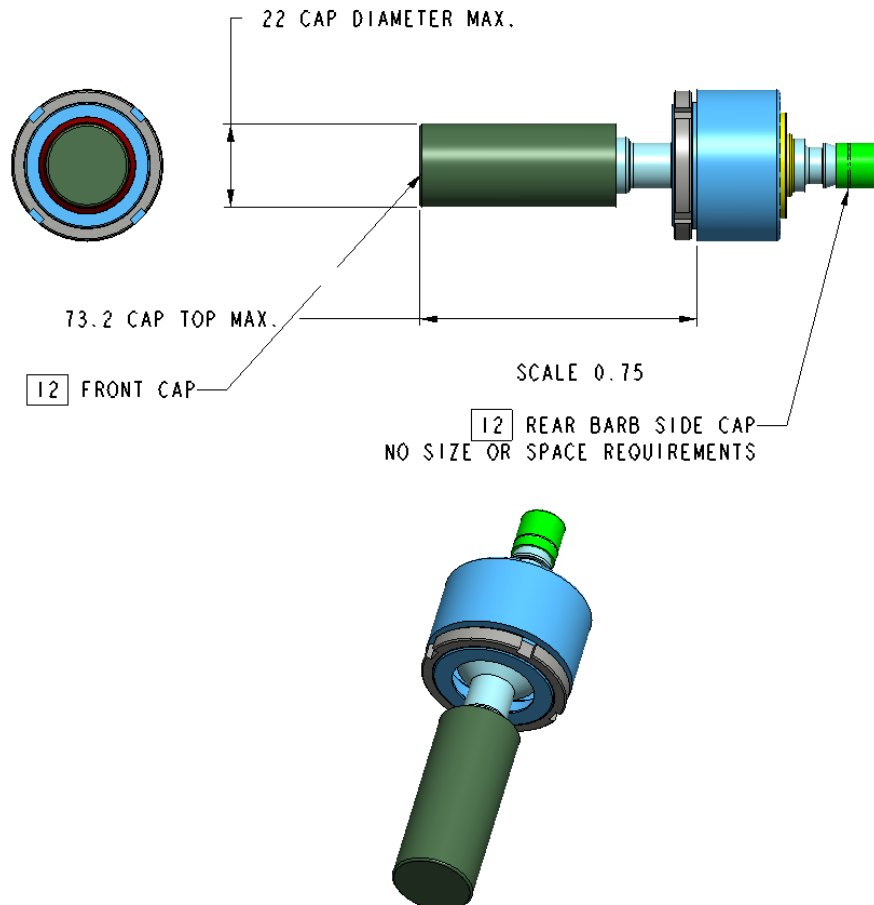


Figure 8.2.2 Plug with Dust Caps

8.2.3 Plug Components - Floating Body

As mentioned in Section 8.2.1, During development, some suppliers within the group added features in their designs to the rear area of the floating body between the barb area and the area that sits in the plug cup to help with self-centering. See section 8.2.1 for implications on space considerations. The detailed dimensions and tolerances on the floating body for these alignment features are marked with a “**” in views below. These features protrude past the main cup depth wise, move with the floating body, and during use can also exceed the diameter of the plug cup during operation at radial and combined

offset conditions due to tolerances and dynamic effects during alignment to mate. The recommendation on future designs to avoid any features that exceed the cup diameter to minimize clearance space around the cup area.

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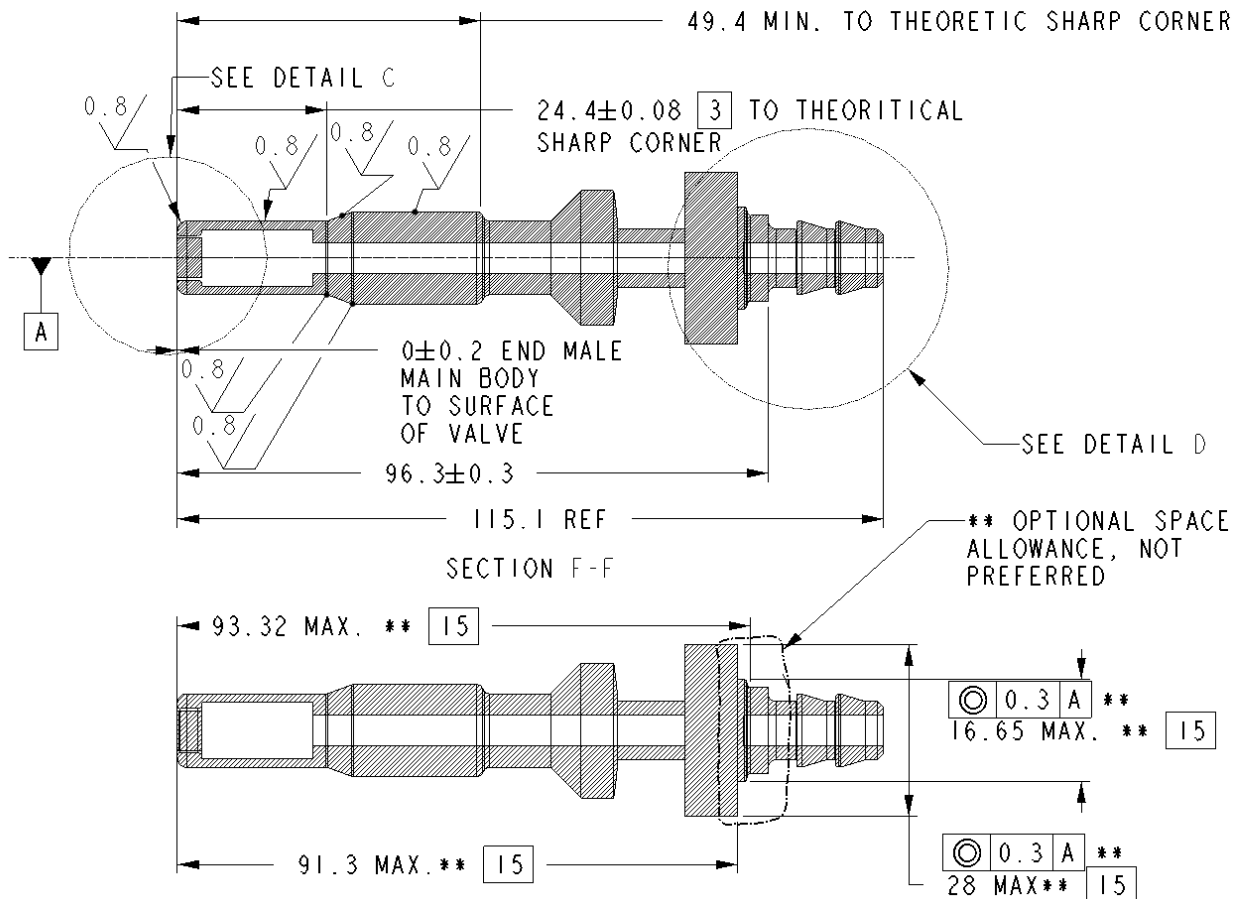
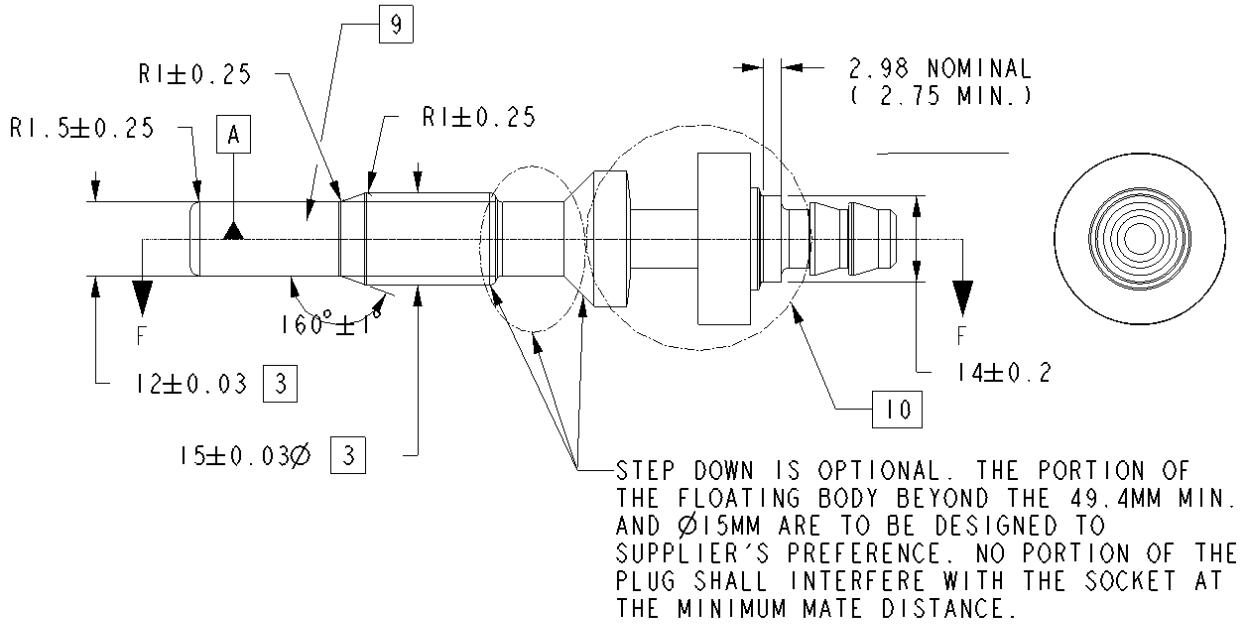


Figure 8.2.3.1 Plug Floating Body Views

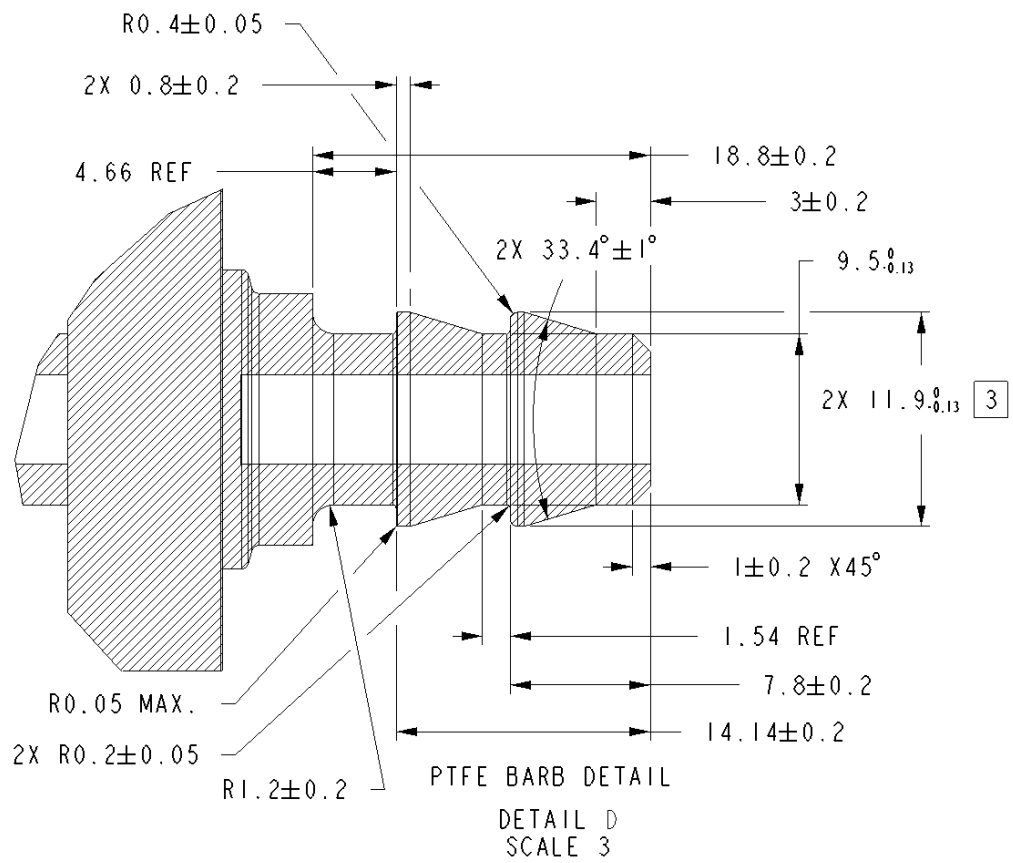
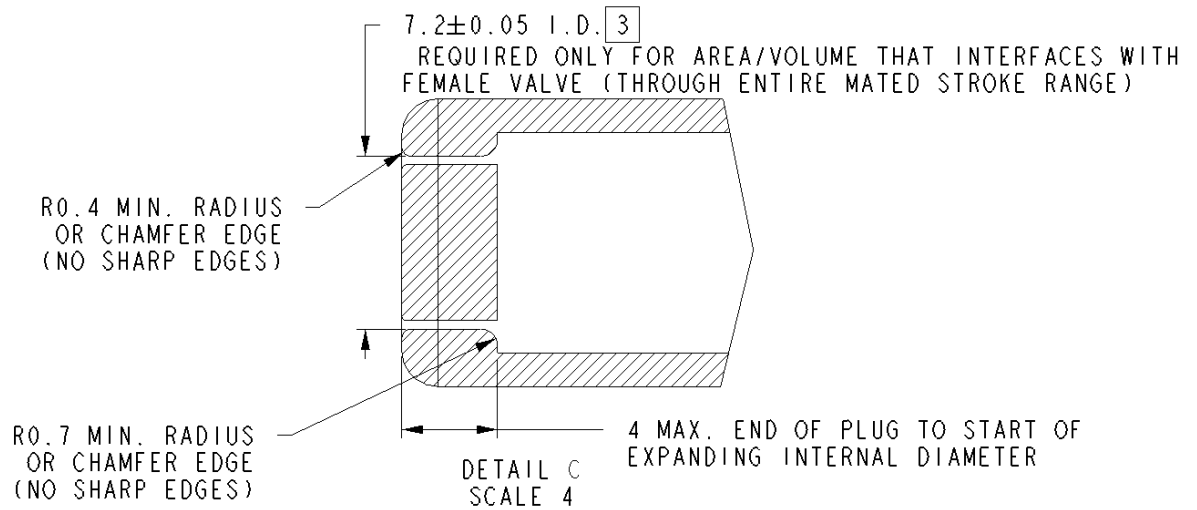


Figure 8.2.3.2 Plug Floating Part Detailed Views

8.2.4 Plug Components - Plug Body “Cup”

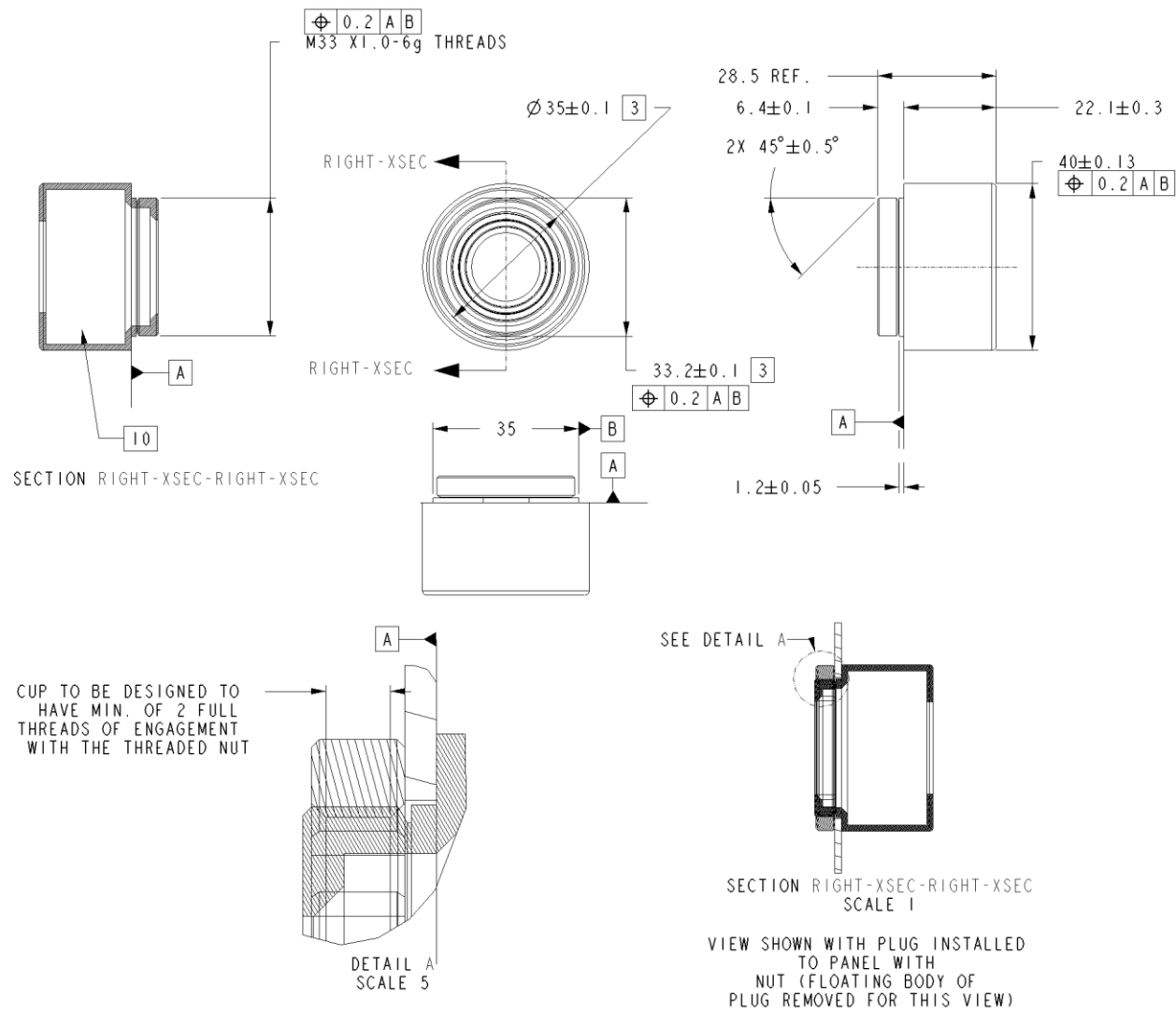


Figure 8.2.4 Plug “Cup” Body Details

8.2.5 Plug Components - Panel Nut

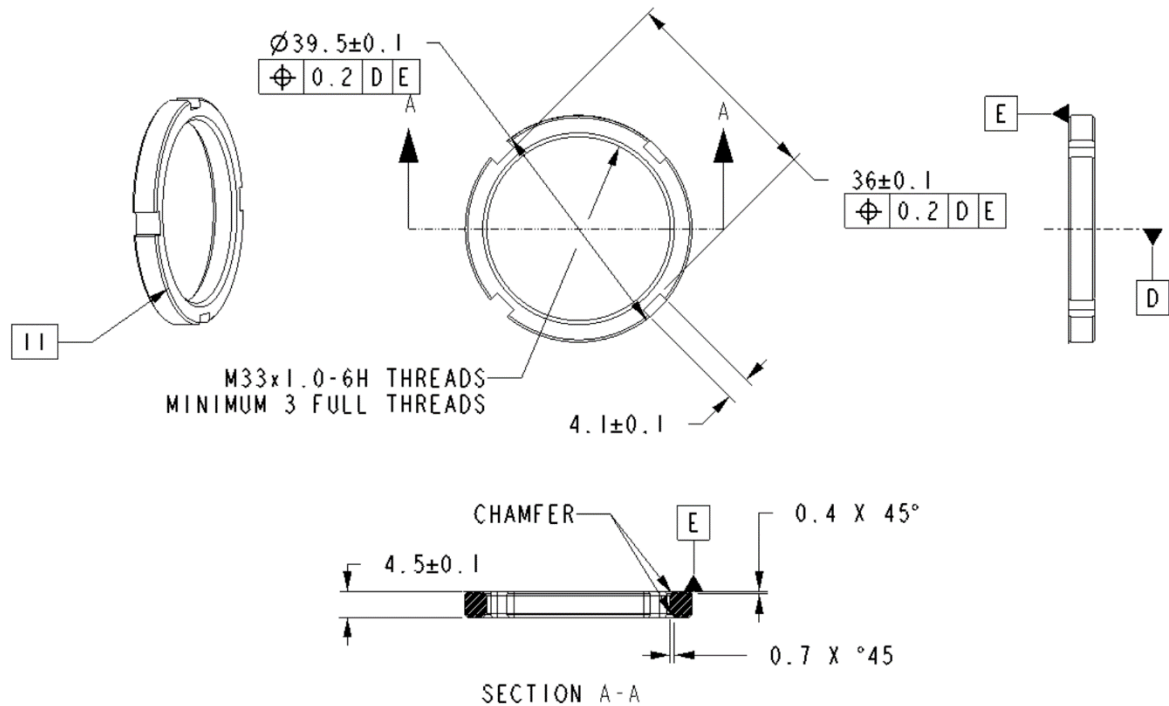


Figure 8.2.5 – Plug Panel Nut Design Information

8.2.6 Plug Weight

Table 8.2.3 Plug Assembly Weight

Maximum Plug Valve Assembly Weight with Panel Nut (excluding dust caps)	189 g
---	-------

9. Functional Parameters

9.1 Misalignment

The design shall meet the allowable misalignment requirements smoothly without binding and meet the force requirements listed in this specification. The design shall also have no change in pressure drop when mated in any offset position within the functional misalignment range.

The requirements below for the valve allowable misalignment are based on the combined effects of both lateral and vertical tolerances, gaps between components,

deflections from loads & weight, and CTE effects. For more details, refer to Section 14.2.2 “Valve Misalignment” in the ORv3 Meta Frame Specification.

9.1.1 Radial-Only Misalignment

The maximum allowable radial-only offset is outlined in figure 9.1.1

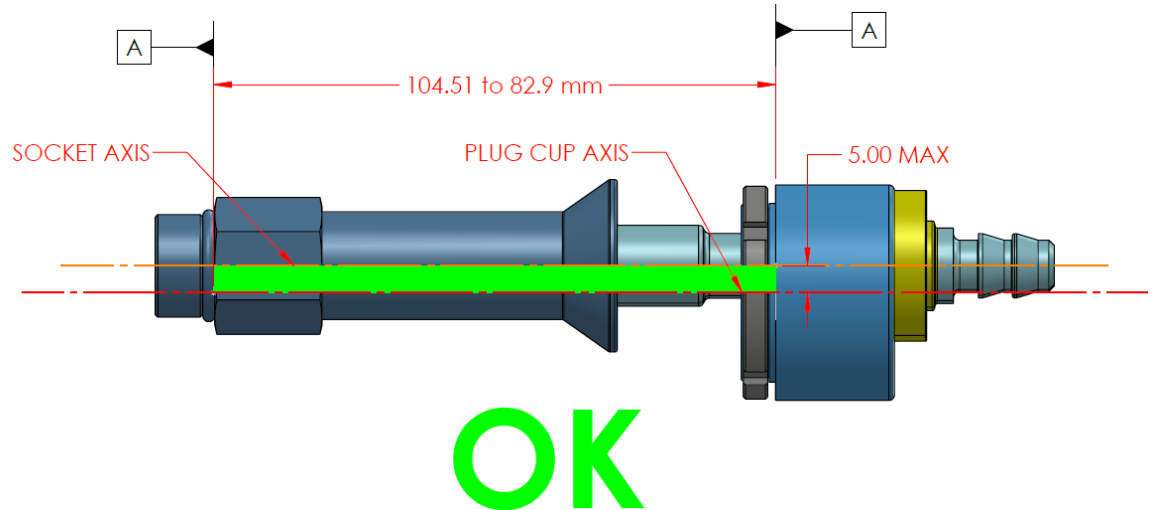


Figure 9.1.1 Maximum allowable pure radial offset. Offset is permissible in any radial direction from the cup axis.

9.1.2 Combined Radial and Angular Misalignment

When dealing with angular offsets, the complexity around projected distances due to angle offset requires the parameters for angular offset to be constrained by certain limitations. As the angle offset increases from parallel, the allowable offset between the plug center axis and socket center axis/centroid reduces due to these projected distances. The valve designs are such that the valve features for alignment are fully aligned (not fully mated) at mate distance of 104.51mm. This is an important distance and is used to define the parameters for allowable combined radial and angular offset.

The self-centering ability of the blind mate plug is provided by the components in inside the plug cup that help center the floating body of the plug. The floating body moves both radially and angularly, but its axis cannot float past 5mm relative to the plug cup axis as measured at Plug Datum A. This allows for up to 5mm of pure radial offset between the socket axis and plug cup axis. The design also allows for up to 2.74° of angular offset. However, when combining radial and angular misalignment there are limitations due to projected angles and distances. As a result, the center axis of the socket projected to the plug Datum A shall never exceed 5mm at the 104.51mm mate distance. A more detailed discussion with some examples is discussed below.

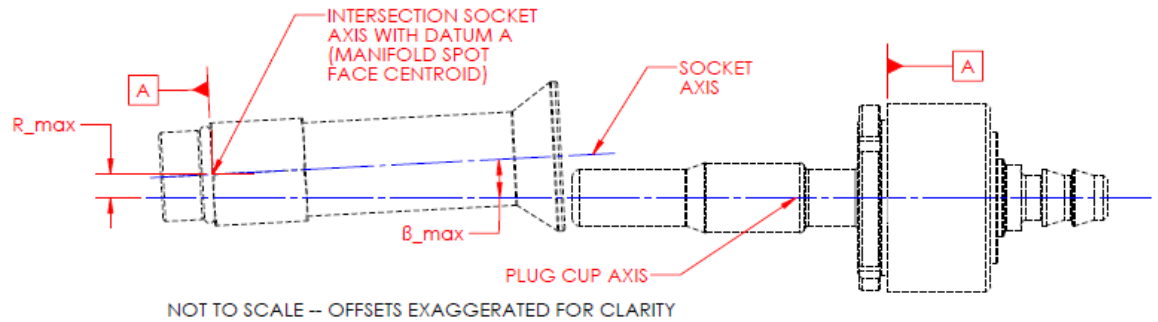


Fig 9.1.2.1 – Layout: critical factors to determine max combined angular & radial offset.

Misalignment capability for the valves is defined by the factors specified in Fig 9.1.2.1. Max radial misalignment, R_{max} , is measured from the theoretical axis projected from the plug cup to the vertex defined by the intersection of the socket axis with socket Datum 'A.' Similarly, max angular misalignment, β_{max} , is defined by the angle created between the socket axis and plug cup axis. Pure radial offset shall never exceed 5mm, and angular misalignment shall never exceed 2.74° . Further, as radial offset increases, allowable angular misalignment is reduced to ensure the socket axis never projects outside of the green zone in Fig 9.1.2.2. The relationship between R_{max} and β_{max} is defined in Formulas 9A and 9.B. Conforming to these maximum values ensures that the floating body of the plug never exceeds the allowable 5mm radial movement under any circumstance. Violation of this limit may result in binding, high forces, or damage to the valves. Reference examples of permissible R_{max} given β_{max} , or vice versa, are provided in Table 9.1.2.

Figure 9.1.2.2 below illustrates an assembly at pure angular offset of 2.74° , resulting in 5mm radial movement of the plug floating body measured at plug cup Datum 'A'. The green region represents an acceptable zone of socket axis radial and angular offset as full alignment between socket and plug is achieved at indicated mate distance.

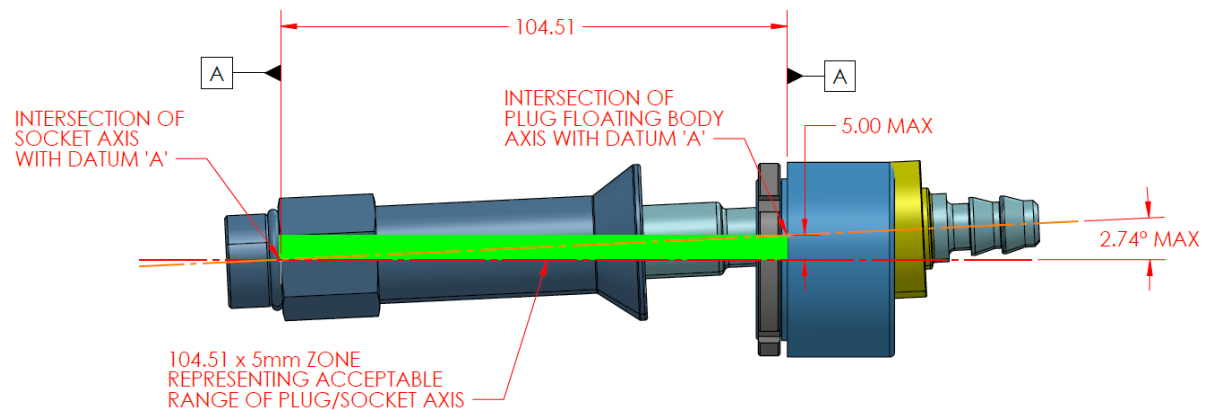


Fig 9.1.2.2

$$R_{\max} = 5 - 104.51 \cdot \tan(\beta_{\max}) \quad \text{Equation 9A}$$

$$\beta_{\max} = \tan^{-1}((5 - R_{\max}) / 104.51) \quad \text{Equation 9B}$$

Formulas 9A and 9B define the relationship between $R_{\max}$ and $\beta_{\max}$ in mm, up to 5mm or 2.74°, respectively.

Table 9.1.2

$\beta_{\max}$	-	-	-	-	-	-	0.00	0.50	1.00	1.50	2.00	2.50	2.74
$R_{\max}$	5.00	5.00	5.00	5.00	5.00	5.00	5.00	4.09	3.18	2.26	1.35	0.44	0.00

Table 9.1.2 - Example permissible $\beta_{\max}$ and $R_{\max}$ values. If $R_{\max}$ is known, reference this table to find the corresponding $\beta_{\max}$ value, or vice versa. These values reference the layout provided in Figure 9.1.2.1. Observe that $R_{\max}$ is unidirectional, but the sign of $\beta_{\max}$ is important in determining allowable radial offset.

9.2 Plug Functional Requirements

To support the misalignment and other functional requirements, the plug has the following functional requirements. The requirements call out motion capabilities along with required dimensions and tolerances. You may note that the radial and combined motion requirements listed in the figures below exceed the requirements needed to meet the misalignment requirements in Sections 9.1.1 & 9.1.2; this is to allow for design margin to avoid functional issues. Any dimensions with ** are for an optional space allowance for centering features; see Section 8.2.3 for details.

9.2.1 Plug Unloaded Requirements

The floating portion of the plug with no load applied shall meet the following design dimensions in figure below. Panel is shown for reference. The spring force required to maintain this distance is outlined in Section 9.4.4.

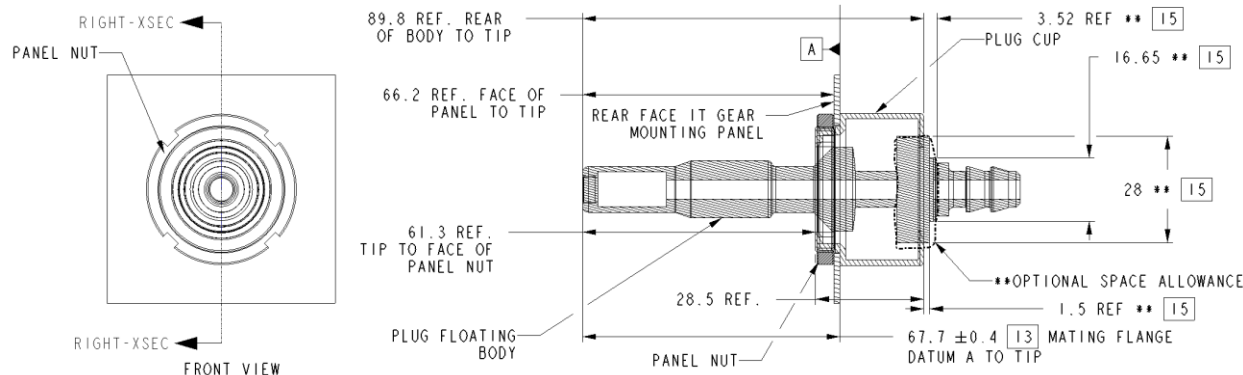


Figure 9.2.1

9.2.2 Self-Centering

The requirements for the plug floating body skew in an unmated state with tubing attached at the barb fitting is shown in Figure 9.2.2 below (tubing not shown for simplicity). The plug floating body shall stay within the zone when:

- (1) in normal state when unmated, unloaded with tubing installed
- (2) after mating and un-mating (floating body must lie within this zone afterwards)

The valves are designed to allow for the misalignment outlined in sections 9.1.1 & 9.1.2. It is critical that the final solution including internal tube routing ensure the floating body of the plug is within this diametrical zone. Failure to meet this requirement may result in increased mate forces, failure to align and mate, binding, and irreparable damage to the valves. A simple jig design was created by the team to verify this during testing; the 3D (step format) CAD is shared in the collateral with this specification. See Section 10 for more details on IT gear internal tube routing.

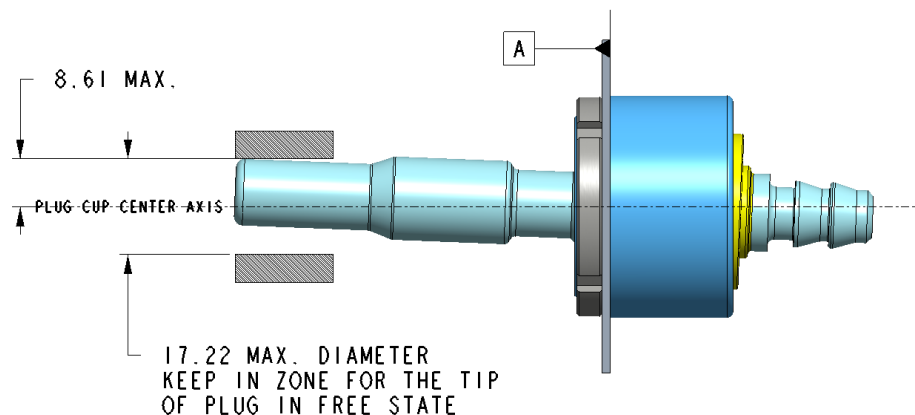


Fig 9.2.2 Plug Allowable Skew in Unloaded State

9.2.3 Plug Loaded General Requirement

When enough force is applied to the floating body on the plug assembly, it moves into the panel towards the rear of the cup (left to right in the view below) until it reaches the internal cup stop at a distance 4.5mm distance from free state. Figure 9.2.3 shows the plug when it reaches the internal stop, and the floating body is radially aligned to the cup. The floating part must also maintain the distance of 63.2mm for the floating part protrusion distance from Datum A within the specified tolerances at the centerline axis of the floating part throughout its motion. The spring force required to push the floating body to the internal stop is outlined in Section 9.4.4.

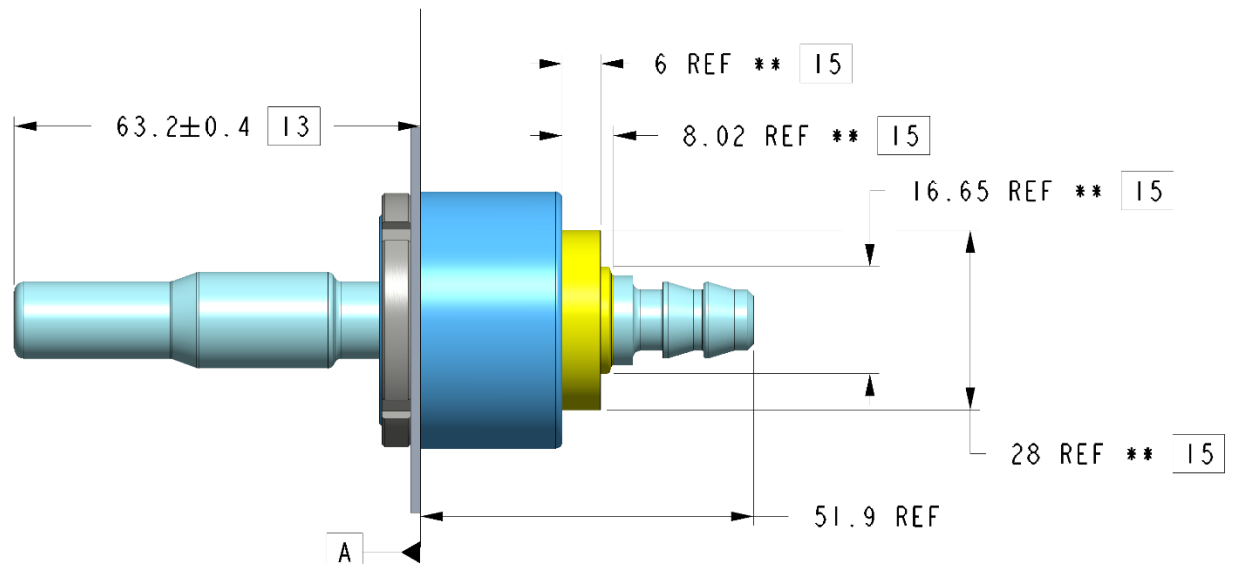


Figure 9.2.3 – Plug Loaded Dimensional Requirements

9.2.4 Plug Radial Motion Requirements

In the loaded state, with the plug pushed to the internal stop in the cup, the plug assembly must meet the requirements for radial only motion in figure below. All views and dimensions below are when the floating portion of the plug is pushed into the cup internal stop. The plug must float a minimum float radially in all directions about the plug center axis as shown in zone shown below. The floating part must also maintain the distance of 63.2mm for the floating part protrusion distance from Datum A within the specified tolerances at the centerline axis of the floating part throughout its motion. Note that requirements below exceed the allowable misalignment listed in section 9.1.1 to allow for fixture tolerances and additional design margin.

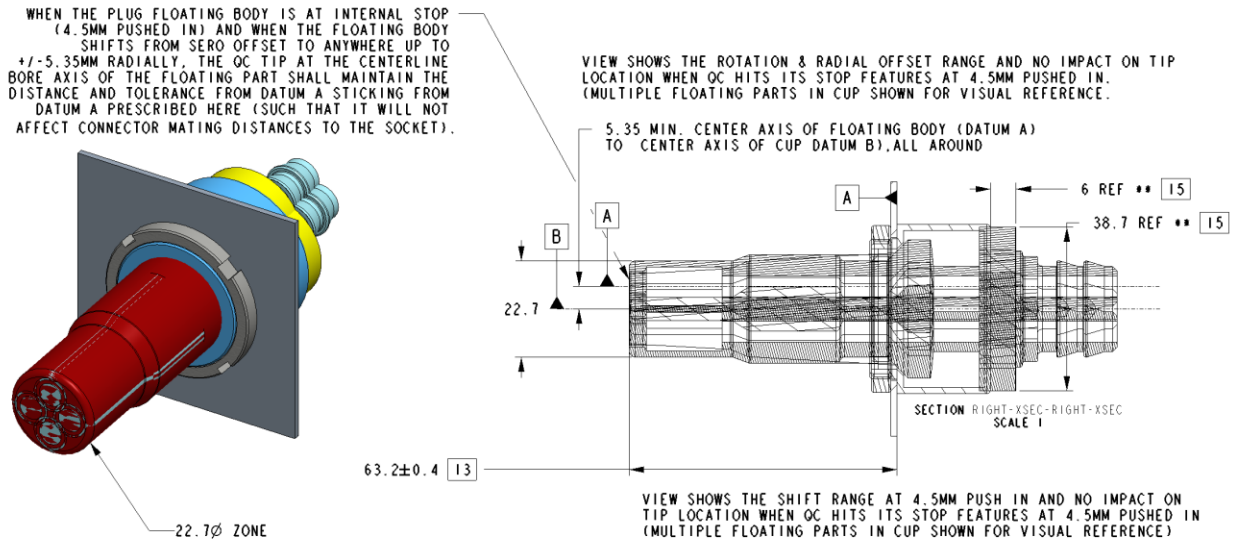


Figure 9.2.4 – Plug Radial Float Requirements

9.2.5 Plug Combined Motion Requirements

All views and dimensions below are at the state when the floating portion of the plug is pushed into the cup internal stop. The floating portion of the valve must float a minimum radially and/or be able to rotate angularly in a minimum and angular zone as shown in the Figure below. The floating part must also maintain the distance 63.2mm floating part protrusion distance from Datum A within the specified tolerances at the centerline axis of the floating part throughout its motion. Note that requirements below exceed the allowable misalignment listed in section 9.1.2 to allow for fixture tolerances and additional design margin.

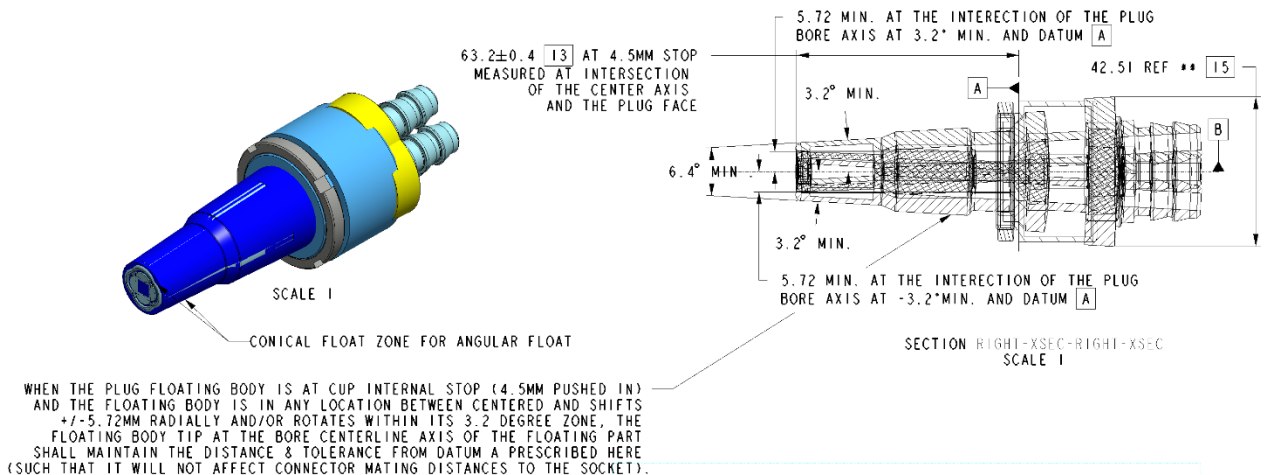


Figure 9.2.5 – Plug Combined Float Requirements

9.3 Working Range:

The working range of the valves is the mate distance within which the mated valves will support full fluid flow. The working range was developed to account for tolerances, gaps between components, and anticipated component deflection from mate loads. See Figure 9.3 and Table 9.3 below for details.

In the ORv3 reference design, when the blind mate liquid cooled chassis is fully installed in the rack and the injectors are flush to the datum A catch edge of the frame, the nominal mate distance from the socket valve Datum A (coplanar with the manifold spot face) to the inner chassis plug mounting surface (plug valve datum A) is 87.4mm referencing Figure 9.3.

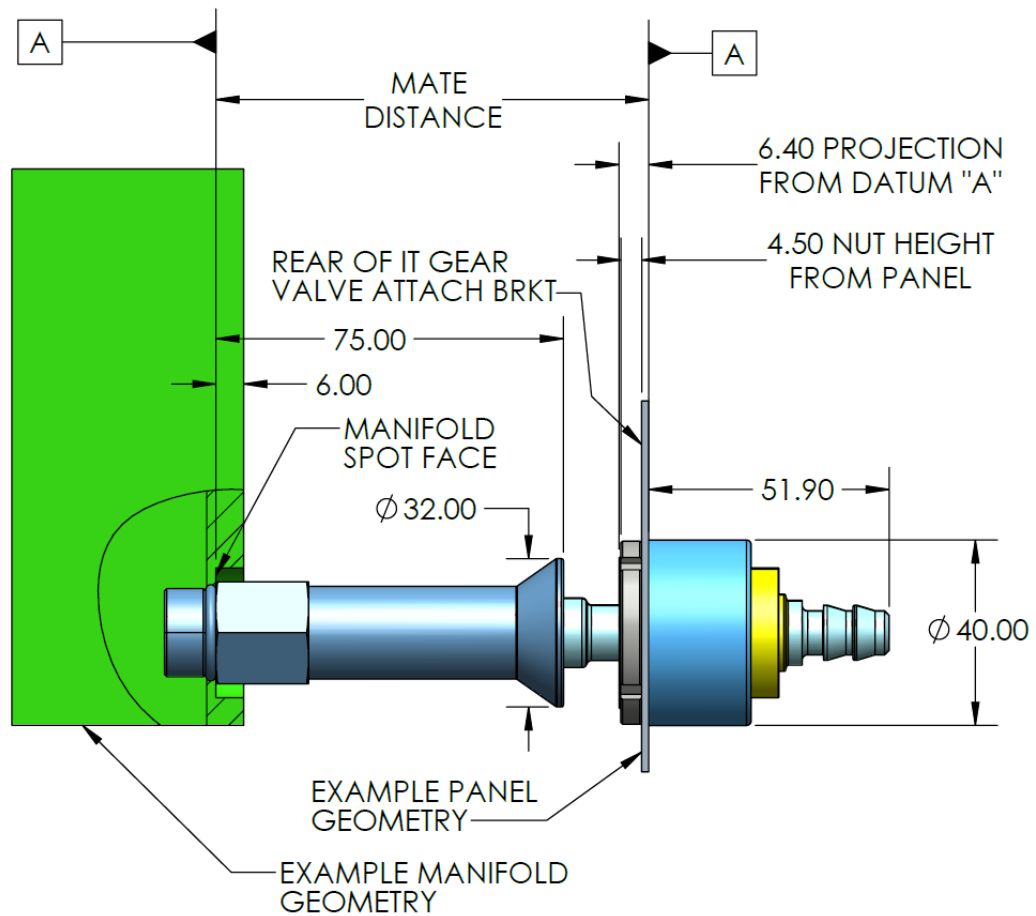


Figure 9.3

Table 9.3 Valve Mate Distance Working Range

Mate Distance “MD” Maximum (mm)	Mate Distance “MD” Minimum (mm)
93.4	82.90

9.4 Mechanical Force Requirements:

The force requirements for the valves in various conditions are provided in the figures below. Having consistent forces for plug and socket ensures that when designs are mixed between suppliers, the forces do not exceed desired maximum values. The development team wanted to define forces at the various distances to help IT gear developers understand the valve mate forces when designing chassis level injector mechanisms used to mate the chassis plug valves to the socket valves (mounted to the manifolds in the rack).

Note that the default routing configuration for the hose, guide, and internal connection with PTFE tubing were used for testing and developing the curves below. Alternate routing layouts, different tubing types and lengths may affect the forces and would need to be tested by developers.

9.4.1 Maximum Static Mate Force

The static mate force requirements are provided below in the default tube routing condition with fluid at 50PSIG. See Section 11 for test details. This must be met in aligned, radial offset and combined radial and angular offset. The mate distance represents the mating distance between Datum A on the socket and Datum A on the plug (ref. Figure 9.3).

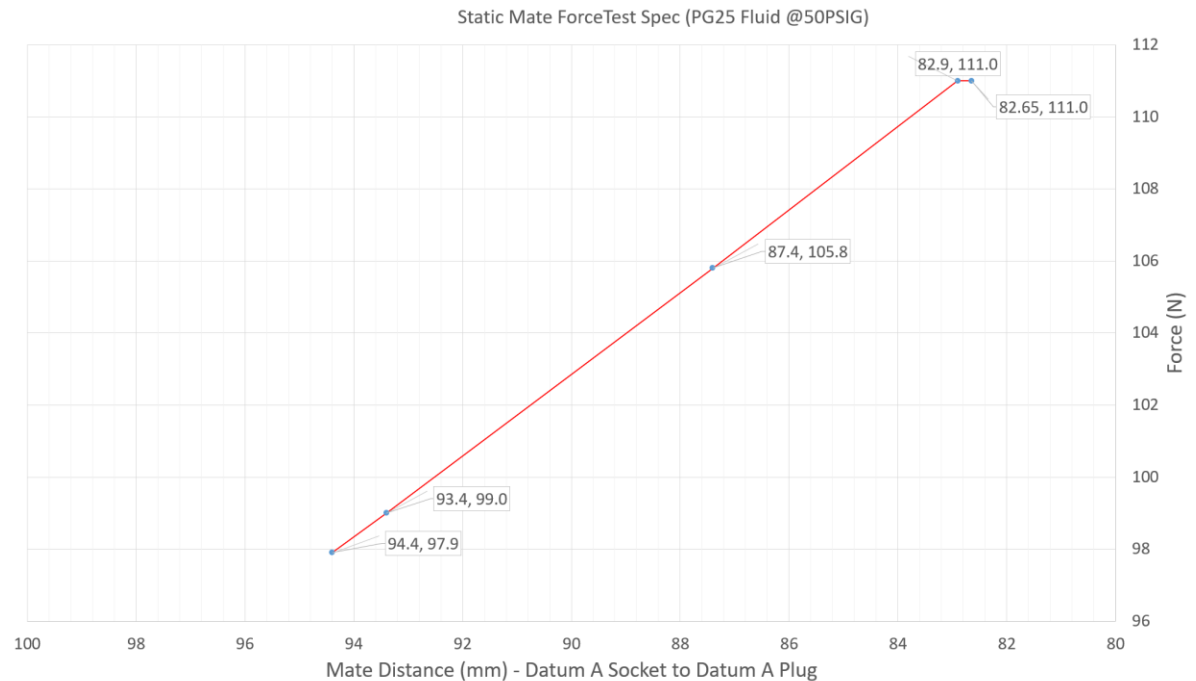


Figure 9.4.1 Max. Static Force @ 50PSIG Fluid Pressure (force vs. Mate Distance)

To ensure reasonable interoperability force variation, there are requirements for the plug and socket spring forces per component as listed below in the table. Note additional points beyond the working range in Section 9.3, +1mm above stage 5 (94.4mm) and -

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0.35mm less than stage 7 (82.65), are listed in the graph as well for test margin (no specific spring force requirements at those mate distances).

Item	Mate Distance (mm)				
	94.4	Stage 5 93.4	Stage 6 87.4	Stage 7 82.9	82.65
Socket Spring Force (N)	NR	27	31.6	35	NR
Plug Spring Force (N)	NR	17	19.2	21	NR
Fluid Pressure @ 50PSIG	39	39	39	39	NR
Friction Force (N)	NR	16N	16N	16N	NR
Total Force (N)	97.9	99	105.8	111	111

9.4.2 Maximum Static Mate Force (Dry)

The dry static mate force requirements are provided below. See Section 11 for test details. This must be met in aligned, radial offset and combined radial and angular offset in the default tube routing condition outlined in the fixture guideline. The mate distance represents the mating distance between Datum A on the socket and Datum A on the plug (ref. Figure 9.3). Note additional points beyond the working range in Section 9.3, +1mm above stage 5 (94.4mm) and -0.35mm less than stage 7 (82.65), are listed in the graph as well for test margin (no specific spring force requirements at those mate distances).

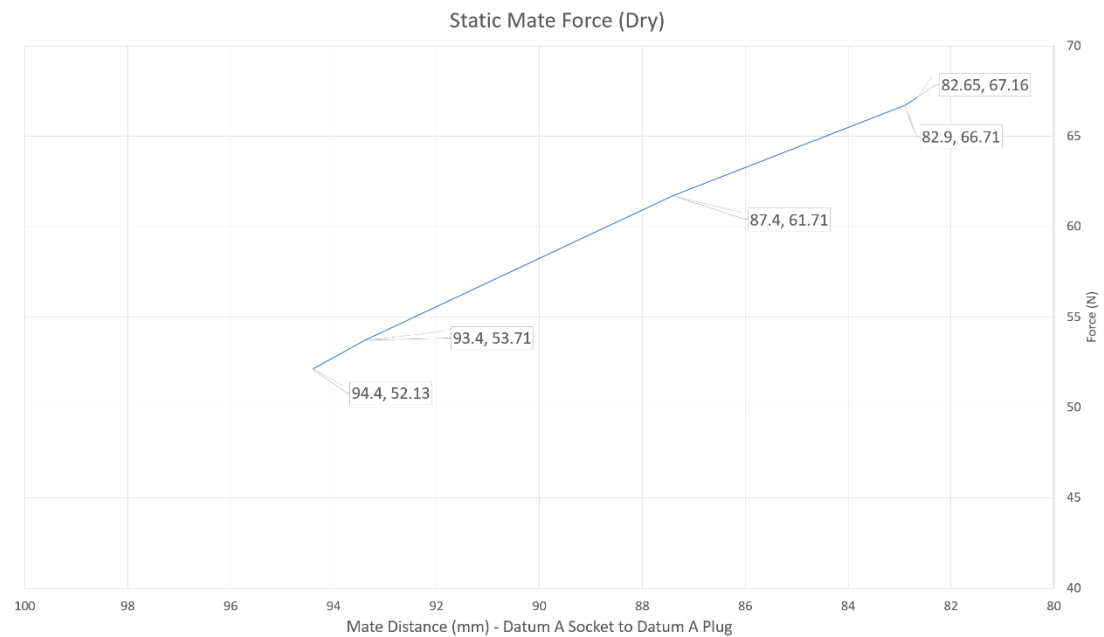


Figure 9.4.2 Max. Static Force Dry (force vs. Mate Distance)

9.4.3 Maximum Dynamic Mate & De-mate Forces

The mate and de-mate force requirements at 40mm/s mate speed with fluid are provided below. See Section 11 for test details. This must be met in aligned, radial offset and combined radial and angular offset in the default tube routing condition outlined in the fixture guideline. The mate distance represents the mating distance between Datum A on the socket and Datum A on the plug (ref. Figure 9.3). Note that during testing an average speed of 0.038 m/s is maintained and during the mate cycles, there is a deceleration to the minimum mate distance, it stops and then accelerates as the pair fully de mates. The deceleration at the end of stroke and acceleration during the de-mate stroke is $5000 \text{ mm/s}^2 \pm 10\%$.

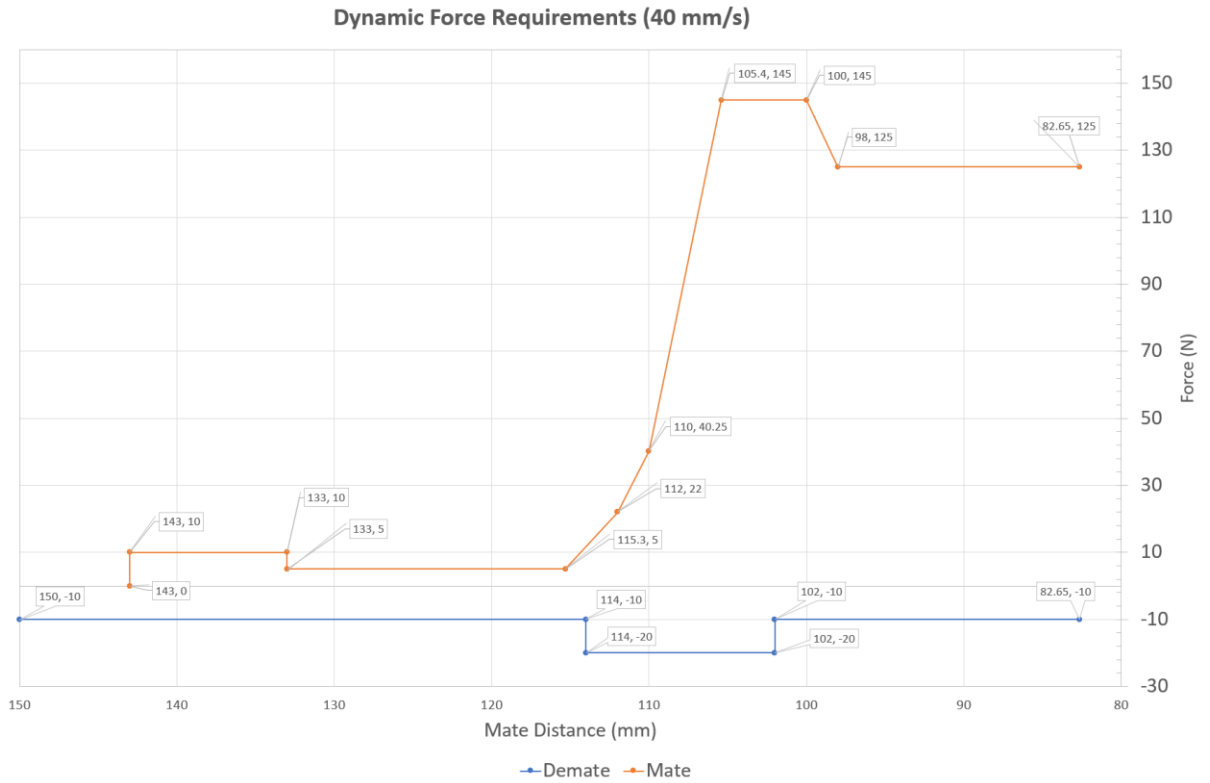


Figure 9.4.3 Max. Dynamic Forces vs. Distance (w/PG25 & 50PSIG)

9.4.4 Plug Centering Spring Force

The plug assembly must meet the following force requirements with no tubing attached at centered condition when floating body is pushed in until it hits the internal stop as outlined in Section 9.2.3 Figure 9.2.3

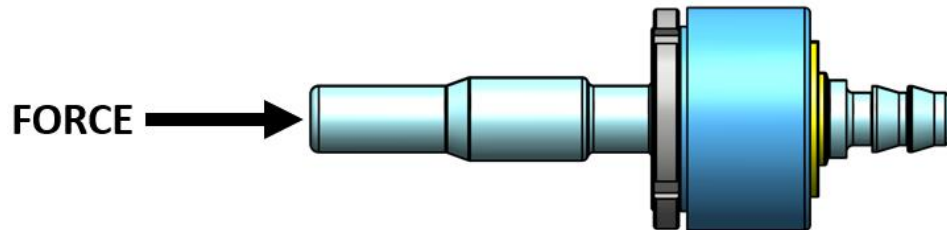


Figure 9.4.4

Table 9.4.4 Plug Centering Spring Force

Parameter	Specification
Spring Force	20N+/- 5N per

9.5 Mate & De-Mate Speed

The maximum mate & de-mate speed for the valve design is 0.31 m/s. The valves shall not exceed maximum force values outlined in Table 11 at this speed. The valves are also tested at a lower speed of 0.038m/s that represents typical mate speed of chassis using the reference injector design mated into the rack.

9.6.1 Durability with Fluid:

The design is to be tested at both .310 and lower 0.038m/s mate speeds. All combination of valves (same and mixed supplier) must meet the following when tested to conditions listed in Table 9.6.1 with fluid:

- A serialized pair of valves must pass 120 cycles tested per conditions listed below in Table 9.6.1 at 0.038 m/s speed and not bind, exceed the mate and de-mate values in Figure 9.4.3, or have any damage.
- Another serialized pair of valves must pass 120 cycles tested per conditions listed below in Table 9.6.1 at 0.31 m/s speed and not bind, have any damage, or exceed the maximum force listed in test 11.1-11 in Table 11.
- Note that testing is performed at slightly higher speeds and with multiple samples to gain confidence in meeting the requirements, but the speeds listed here are the maximum allowed by the user.

Table 9.6.1 Durability (Wet) - Mate Cycle Requirements with Fluid

Condition	# of Cycles
Aligned	40
5mm Radial Offset	40
Combined Radial and Angle Offset	40
	120 total mate cycles

9.6.2 Durability without Fluid:

Any combination of valves (same or mixed suppliers) must meet the following when tested to conditions listed in Table 9.6.2 without fluid:

- Another serialized pair of valves must pass 36 cycles tested per the conditions listed below in Table 9.6.2 at 0.31 m/s speed and not bind, have any damage or exceed the maximum force listed in test 11.1-10 in Table 11.
- Note that testing is performed at slightly higher speeds and with multiple samples to gain confidence in meeting the requirements, but the speeds listed here are the maximum allowed by the user.

Table 9.6.2 Durability (Dry) - Mate Cycle Requirements without Fluid

Condition	# of Cycles
-----------	-------------

Aligned	12
5mm Radial Offset	12
Combined Radial and Angle Offset	12
	36 total mate cycles

9.7 Maximum Fluid Temperature

The overall solution is developed based on a maximum working fluid temperature of 60°C.

9.8 Flow Performance & Other Requirements

Table 9.8: Various Requirements

Parameter	Specification
Working range of Valves	See Section 9.3
Pressure Drop	See Figure Below
Maximum Working Pressure	3.45 bar (50PSIG)
Maximum Fluid Temperature	See Section 9.8
Spillage	0 PSIG: 0.025cm ³ 50 PSIG: 0.05cm ³
Hydrostatic pressure Coupled	150 PSIG
Hydrostatic pressure Un-coupled	150 PSIG
O-Ring Blow Out	55 PSIG
Air Inclusion	0.1 ml/cycle
Maximum Flow Rate	9LPM

9.8.1 Pressure Drop Requirements

The requirements outlined in Figure 9.8.1 below shall be met at any mate distance within the working range outlined in Section 9.3. Note that the testing in Table 11.1.1 was performed at slightly margined values to account for test tolerance, valve manufacturing tolerances, and provide confidence in meeting the requirements. Test conditions are as follows:

- Fluid supply temperature: 40°C

- Fluid: Dowfrost LC25 (25% propylene glycol water mixture)

Figure 9.8.1 outlines the range (lower-to-upper) of acceptable pressure drop for different flow rates. The valves must meet the requirement in both plug-to-socket and socket-to-plug flow directions. This requirement must be met by all vendors intending to supply these parts. Same supplier and interoperability testing with all suppliers and all combinations shall be tested. For more details see Table 11.1 Test 11.1-1

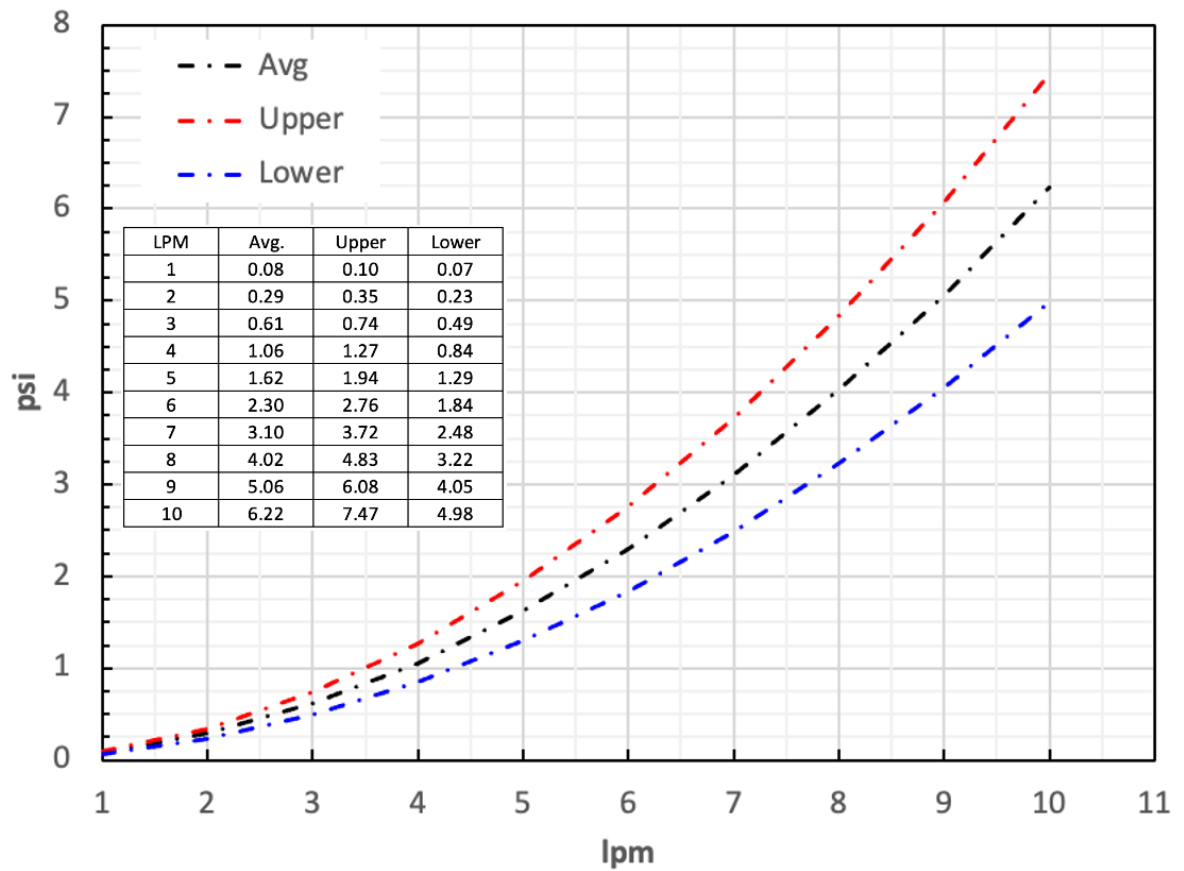


Figure 9.8.1 – Flow Impedance Requirements

10. Tubing Interfaces and Considerations

10.1 PTFE Tubing Info

The fluid conveyance tube used for the blind mate valve solution is a convoluted, single wall, PTFE (Polytetrafluoroethylene) construction. This tube type was specifically chosen as the connection between the valve tip and internal chassis liquid loop for its mechanical, chemical, and routing properties. The team iterated on several versions of this PTFE tube with varying wall thickness, convolution pitch, and barb geometry for optimal performance with the valve. All valve testing conducted by the team and displayed in this specification used this PTFE tube. The length shown in the reference images below is for the default layout configuration as well as what was used in fixtures for testing but can be modified by the end user for their use case. This tube is designed to be pushed onto the barb using an installation tool referenced in Section 7.5.2.

Reference information on the tube is in the figure below:

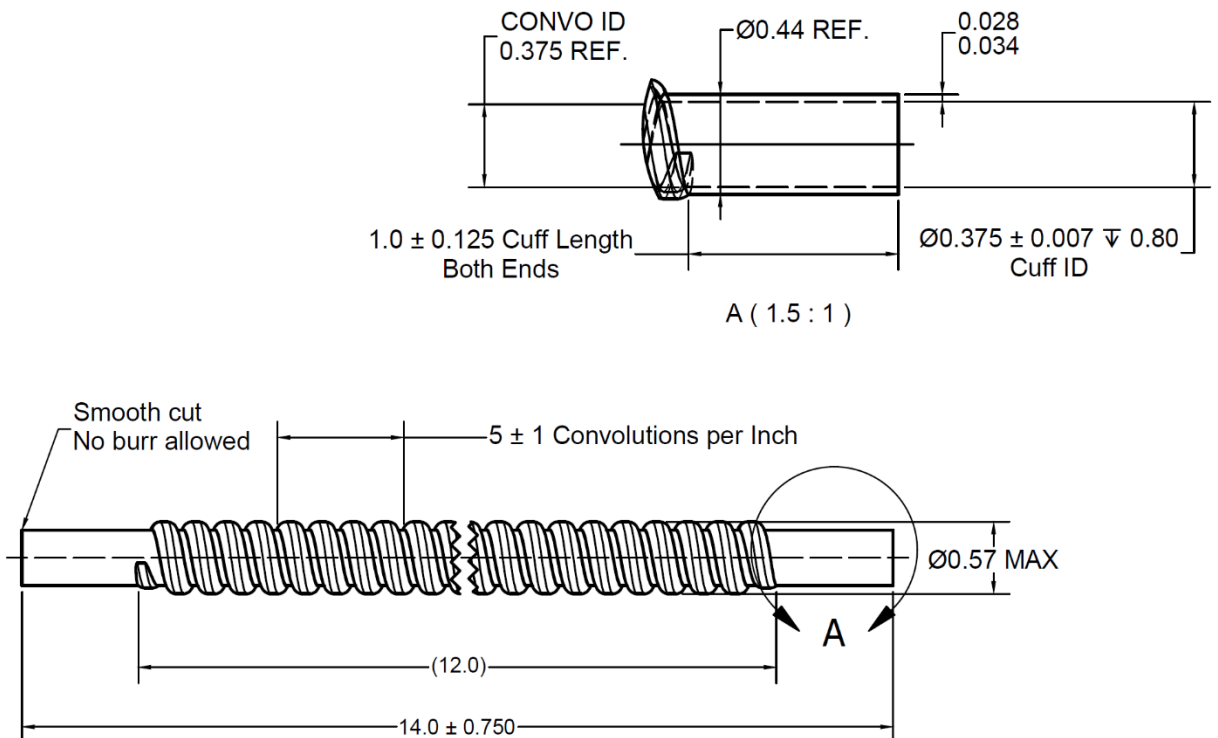


Figure 10.3

10.2 Barb Design Details

The barb on the end of the plug is designed to work specifically with the PTFE tube chosen for this application above. The diameter, number of barbs, and ramp geometry were chosen so that the 3/8" ID PTFE tube with a straight end cuff can be pushed onto the barb and create a pressure tight seal without the use of hose clamps. See Figure 8.2.3.2 for dimensions of this barb. A tube installation tool is recommended for installation of the tube onto the plug, as referenced in the IT Gear Design Guide.

Note the barb geometry shown is for reference and not a requirement. Developers are free to work with suppliers who developed the design to create their own hose & tubing connection geometry, however special care shall be taken as the influence of tubing design can have a significant impact on the blind mate valve mechanical performance.

10.3 Selection & Routing Considerations:

During testing and development, tube material and internal chassis routing significantly influenced the performance of the BMQC valves. Hose or tube stiffness, outer material, and diameter were found to play a large factor in the mate forces required for valve mating and preventing potential binding. For this reason, the above-mentioned PTFE tube was chosen as opposed to other less flexible materials such as rubber hoses.

Other potential tubing solutions & materials would need to be tested and evaluated by the developer. Also, as mentioned in Section 10.2, the barb design would also need to be evaluated and possibly modified by the developer.

Internal chassis layout of the liquid loop and subsequent hose routing also contributed to valve mate forces and a default layout with hose guides was created to minimize the impact of bend radii and tube stiffness on the valve plug. Other chassis layouts are possible but need to be tested and validated by the end user for mating performance. For more detailed design guidelines, please refer to the Open Rack V3 IT Gear Design Guide.

10.4 Hose Guide & Insert Reference Information

To help ensure proper function of the plug floating body while attached to a hose or tube, hose guides were developed and tested at varying distances from the BMQC plug. The goals of the hose guides were to support the weight of a coolant filled tube behind the plug to prevent sagging (and potentially hitting nearby features or components), allow low friction sliding of the tube outer surface in the guide when the plug is pushed in and/or angled, and allow a smooth bend radius if the hose and routing have a bend in it.

A 3D model of this guide is included with this specification as well as the reference image below. For more details, please refer to the IT Design Guide.

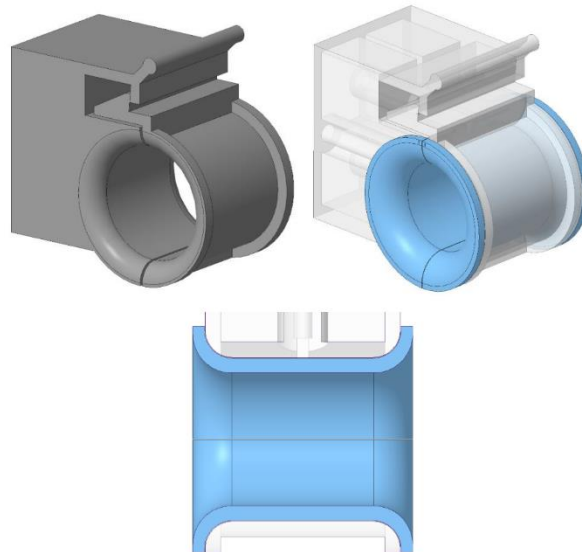


Figure 10.4 Showing Guide (grey) & Insert (blue)

10.5 Tubing Installation on Barb

The PTFE tube developed for the blind mate plug is installed onto the barb end of the plug tip using an installation tool that is like other hose barb levered installation tools, but specifically designed for holding the plug cup and tip firmly while the tube is pushed onto the barb with a lever arm. The design is a reference design and is a prototype. End users and developers should do the appropriate evaluation to determine if the tool works in their application.

The PTFE tubing clamp should be sufficient to grip the tubing for installation but must avoid over clamping which may damage or permanently distort the tube.

During installation of the tubing onto the barbs, make sure to check the tubing after barb insertion for collapse or distortion as it is possible to buckle the tubing if too much force is used or force is applied after it is fully inserted over the barbs. Detailed inspection for all tubing assemblies to make sure the tubing is fully inserted and there is no physical damage to the tube such as kinking, collapse and/or cuts. Excessive deformation, kinks, cuts, and/or other physical damage could lead to leaks or other performance issues.

For the installation design shared, the installer opens the QC Plug clamp and places the plug body inside against the rubber surface. The clamp is closed and latched closed. On the other side, the PTFE tube is placed on the hose clamp rubber surface and latched

closed. The installer uses the rotating handle lever arm to push the tube onto the barb until the tube cuff is flush with the barb shoulder.

A drawing and 3D cad model of the reference installation tool is included in the collateral.

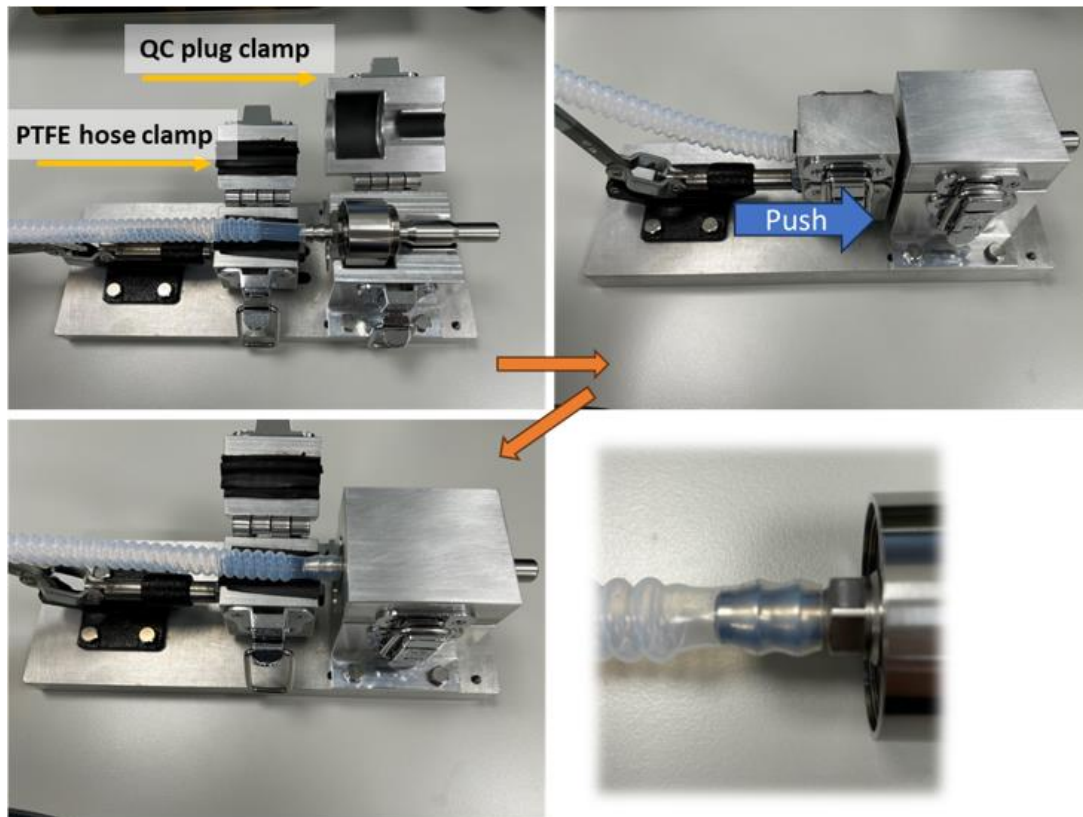


Figure 10.5 PTFE hose installation tool

11. Test Requirements and Methodology:

Below are the test requirements for the valves. Most tests listed below require the testing of valve combinations with same and mixed supplier valves as well as multiple samples of each socket-plug combination. There are also safety test requirements in Section 15. For a design to be considered to have met the requirements specification, the valve design must be tested and pass all tests below including those requiring supplier interoperability. Interoperability testing must be performed with all approved suppliers in accordance with any test in the table below. Test levels below are generally tested with margin above the specifications listed in various other sections of this specification.

Additional Information:

- Unless specified otherwise, all tests listed below are to be executed at accepted ambient temperature and relative humidity levels.
- Pre-conditioning, and functional verification of samples used as part of the test shall be completed.
- Post test, functional verification & inspection of the samples shall be completed on all tests
- The testing that requires the test fixture shall use a fixture manufactured per requirements in Section 14.
- A sample of the test plan and test order the development team used for testing is supplied with the collateral; if additional valves are to be tested the matrix would need to grow to ensure all combinations are tested to the appropriate quantities outlined in table

General Pass/Fail Criteria

The valves must not bind, leak, exhibit damage to external or internal parts, have any degradation in performance beyond the requirements, meet any additional requirements listed in a particular test below. and meet any requirements in any reference test specification listed in “Reference Standard” column below for that test.

Table 11.1 Minimum Test Requirements

Test # & Description	Requirements	Reference Std.
11.1-1 Pressure Drop Test	• Pressure 50PSIG • Ambient Temperature: 20C ± 3C • Fluid Temperature: 40C • Fluid Type: PG25 (see note A) • Must meet Figure 9.8.1 curve at any mate distance between stages 5 (93.4mm) & 7 (82.9mm), but also measure at stage 5 +.4mm (93.8mm) & stage 7 -.4mm (82.5mm) to verify extreme ends of stroke are met when accounting for tolerances of design and fixture • Flow rates: measured at 2, 4, 6, 9, & 10 LPM. • Requirements are listed in Figure 9.8.1 • Same supplier and interoperability testing required, minimum 3 samples of each same supplier and interop combination	ISO 18869:2017, Section 13 Conditions at left supersede any conflict with the referenced standard. Reference Fixture design Guidelines
11.1-2 Air Inclusion	• Pressure: Atmospheric • Ambient Temperature: 20C ± 3C • Fluid Temperature: 20C ± 3C • Fluid: PG 25 or water • Requirements are listed in Table 9.8 • Same supplier and interoperability testing required, minimum 3 samples of each combination	ISO 18869:2017, Section 11

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11.1-3 Spillage	- Pressure: 0 & 55 PSIG - Ambient Temperature: 20C ± 3C - Fluid Temperature: 20C ± 3C - Fluid type: water - Max. spillage: see Table 9.8 for requirements - All performing with vertical orientation and using graduated cylinder per ISO Standard listed - Provide test results for both pressure levels - Same supplier and interoperability testing required, minimum 3 samples of each same supplier and interop combination	ISO 18869:2017, Section 12
11.1-4 Hydrostatic pressure Coupled	- Pressure: 150 PSIG, test to this pressure (not to burst) - Duration: 3 minutes - Ambient Temperature: 50C+3C/-0C - Fluid Temperature: 50C+3C/-0C - Fluid: Water - Tubing: convoluted PTFE reference design - Tube Routing: default aligned condition, See fixture guidelines - Valves shall be allowed to stabilize at 50C for 1 hour to ensure fluid, tubing and valves are at temperature prior to starting pressure test - Same supplier and interoperability testing required, minimum 3 samples of each same supplier and interop combination	ISO 18869:2017, Section 18.3 Reference Fixture design Guidelines Section 14 for fixture layout & design.
11.1-5 Hydrostatic pressure un-coupled	- Pressure: 150 PSIG, test to this pressure (not to burst) - Duration: 3 minutes - Ambient Temperature: 50C+3C/-0C - Fluid Temperature: 50C+3C/-0C - Fluid: Water - Tubing: convoluted PTFE reference design - Tube Routing: default aligned condition, See fixture guidelines - Valves shall be allowed to stabilize at 50C to ensure fluid, tubing and valves are at temperature before testing - Same supplier, minimum 3 samples	ISO 18869:2017, Section 18.2 Reference Fixture design Guidelines Section 14 for fixture layout & design.
11.1-6 O-ring Worst case	- Pressure: 55 PSIG (one side), 0 psi on opposite side, test both directions. - Ambient Temperature: 20C ± 3C - Fluid Temperature: 50C - Fluid: Water - Tubing: convoluted PTFE reference design - Tube Routing: default aligned condition, see fixture guidelines fixture set-up for this test - Set-up: Assemble the socket and plug in the test fixture, set the speed of the test speed to 40 mm/s. Connect the fluid line to the socket with pressure listed above. Connect the plug side to tank or open drain. Make 10 connections and check if O-ring is still in correct position. Repeat the test with pressure on the plug side and socket connected to tank or open drain. - Same supplier, minimum 3 samples	Reference Fixture design Guidelines Section 14 for fixture layout & design.
11.1-7	- Pressure: N/A (Dry) - Ambient Temperature: 20C ± 3C - Fluid: N/A (Dry)	N/A

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Plug Centering Spring Force, (Dry)	• Test Conditions: installed in panel or fixture, without tubing installed • Measuring force required to make floating portion of valve reach the internal rear stop at 63.2mm per Figure 9.5. • Must meet force requirement in Table 9.7.4 • Same supplier only, minimum 3 samples	
11.1-8 Static Force vs. Mate Distance, Mated Pair, (Dry)	• Pressure: N/A (Dry) • Ambient Temperature: 20C ± 3C • Fluid: N/A (Dry) • Tubing: convoluted PTFE reference design • Tube Routing: default aligned condition, see fixture guidelines fixture set-up for this test • Mating conditions: 1. Aligned 2. Radial: 5.25mm offset 3. Combined: 3 deg & 5.25mm radial • Measure Static force at Mate Distances 94.4, 93.4 (Stage 5), 92.4, 91.4, 90.4, 89.4, 88.4, 87.4 (Stage 6), 86.4, 85.4, 84.4, 83.4, 82.65 (Stage 7 with margin), in the exact order listed and stopping at each stage, with each plug & socket pair • Dwell at each distance for 1 sec • Must not exceed mate forces in Figure 9.4.1 • minimum 3 samples of each same supplier and interop combination	Reference Fixture design Guidelines Section 14 for fixture layout & design.
11.1-9 Static Force vs. Mate Distance, Mated Pair, (Wet)	• Pressure: 50PSIG (both sides pressurized for tests) • Ambient Temperature: 20C ± 3C • Fluid Temperature: 50C • Fluid: PG 25 • Tubing: convoluted PTFE reference design • Tube Routing: default aligned condition, see fixture guidelines fixture set-up for this test • Mating Conditions: 1. Aligned, 2. 5.25 radial offset 3. 5.25 radial offset & 3° angle offset • See fixture guidelines section for fixture set-up for this test • Measure Static force at Mate Distances 94.4, 93.4 (Stage 5), 92.4, 91.4, 90.4, 89.4, 88.4, 87.4(Stage 6), 86.4, 85.4, 84.4, 83.4, 82.65 (Stage 7 with margin), with each plug & socket pair • Dwell at each distance for 1 sec • Must not exceed mate forces in Figure 9.4.2 • Same supplier and interoperability testing required, minimum 3 samples of each same supplier and interop combination	Reference Fixture design Guidelines Section 14 for fixture layout & design.
11.1-10 Mate Cycles, Mated Pair, (Dry)	• Pressure: 0 PSI (no fluid), PG25 post-test check • Ambient Temperature: 20C ± 3C • Fluid Temperature: not applicable (dry) • Fluid: not applicable (dry) • Tubing: convoluted PTFE reference design • Tube Routing: default aligned condition, see fixture guidelines fixture set-up for this test	Reference Fixture design Guidelines Section 14 for fixture layout & design.

	• Velocity: Run at 330 mm/s $\pm$ 40mm/s • 12 cycles per set-up below (3 tests) • Alignment conditions: 1. Aligned 2. Radial: 5.25mm offset 3. Combined: 3 deg & 5.25mm radial • Must not exceed 266N during any portion of mate/de-mate cycle. • A mate cycle is described as follows: starting from completely unmated to 82.65 (Stage 7 with margin) back to completely unmated • Same supplier and interoperability testing required, minimum 3 samples of each same supplier and interop combination • Upon completion of above, valves to be tested with fluid @ 50PSI to ensure no leaks or functional issues (both unmated and mated) • Plug valve shall be checked for compliance to self-centering zone shown in figure 9.2.1.2 in prior to starting, during test and after last cycle	
11.1-11 Mate Cycles, Mated Pair, (Wet)	• Ambient Temperature: 20C $\pm$ 3C • Fluid Temperature: 50C • Pressure: 50 PSI (both sides pressurized for tests) • Fluid: PG25 • Tubing: convoluted PTFE reference design • Tube Routing: default aligned condition, see fixture guidelines fixture set-up for this test • Velocity: Run at (1) 40mm/s $\pm$ 2mm/s and (2) 330 mm/s $\pm$ 4mm • A mate cycle is described as follows: starting from completely unmated to 82.65 (Stage 7 with margin) back to completely unmated • 240 total cycles, 120 per speed, 40 per alignment condition • One serialized pair to be used for each 120 mate cycle velocity run • Alignment conditions: see details below 1. Aligned: 40 cycles at each speed 2. Radial: 5.25mm offset, 40 cycles at each speed 3. Combined: 3 deg & 5.25mm radial, 40 cycles at each speed 4. Measure force and develop curve for mate & de-mate for at least 1 cycle per condition. • Same supplier and interoperability testing required, minimum 3 samples of each same supplier and interop combination • Plug valve shall be checked for compliance to self-centering zone shown in figure 9.2.2 in prior to starting, during test and after last cycle • For 40mm/s test 1. measure mate & de-mate forces during test at a minimum of cycle #1 and #40 for each condition, 40 cycles per condition totaling to 120 cycles; no damage, binding or leaks, must meet mate and de-mate curves in Figure 9.7.3 for all test cycles. 2. Must not exceed mate forces in Figure 9.4.3 for the	Reference Fixture design Guidelines Section 14 for fixture layout & design.

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	- For 330 mm/s tests: - no binding, pass all functional tests during and after completion, and show no sign of damage or leaks. No force measurements requirement. - Must bind or exceed 266N during testing 330 mm/s	
11.1-12 Plug Centering Verification	- Supplier created tool that replicates the keep in zone in section 9.2.2 - Standardized, 3D printed tool files provided for visual measurement of offset to the nearest 1mm. - Tubing: convoluted PTFE reference design (1) move plug valves tips angle & radial) in at least 4 orthogonal directions and verify part returns to accept zone with tubing connected and routed (2) check centering after mate cycle testing in each of the 3 mate conditions - Valve shall return to be within zone outlined in Figure 9.2.1. - See fixture guidelines section for fixture set-up for this test - All centering verification to be performed using a go/no-go gauge based on the design included with the design collateral. The gauge replicates the keep-in zone self-centering requirements in figure 9.2.2 in the specification. During the measurement, no part of the tip of the floating body shall touch the gauge tool. - Minimum 3 samples, same supplier	Reference Fixture design Guidelines Section 14 for fixture layout & design.
11.1-13 Torque, Component	- Perform torque test on the socket into manifold port at 120% of the maximum value listed in table 7.6.1, for material and port design reference manifold spec., For valve, no stripping of threads, no damage to socket interfaces, and no leaks allowed. - Perform torque test on plug assembly with nut to 120% of the maximum value listed in table 7.6.1 using the tool design provided in this specification. No deformation of panel and no stripping of threads, no damage to plug or nut interfaces, and no damage to nut install features allowed Notes: - only the plug body and sheet metal made per panel design requirements listed in this specification are to be used to constrain the plug during test. - Tester may need multiple sheet metal panels - Minimum 3 samples, same supplier	
11.1-14 Plug Float & Position Verification	- Minimum 3 samples, same supplier - Reference Figures 9.2.3 & 9.2.4 - Apply load to make plug floater reach internal stop - Verify plug floats radially in any direction radially in zone including centered position provided in figure 9.2.3 and 63.2mm dimension is maintained within tolerance. - Verify plug floats and rotates ("combined") in any direction in zone provided in figure 9.2.4 and 63.2mm dimension is maintained within tolerance.	
11.1-15 External Force	One sample.	UL62368-1 2 nd edition

	• Tubing: convoluted PTFE reference design • Apply the pulling force of 30N (+/- 10%) for 5 seconds in any direction on a hose accessible by an operator during normal operation and servicing. Apply 30N force in any direction to the hose connected to plug or socket (both mated and unmated). During and after the test, there shall not be rupture or leakage. A pull force gauge can be used for this testing.	
11.1-16 Packaging	• (2A OR 2B) Single Parcel 1. For packages <= 150lbs follow Section 2A 2. For packages > 150lbs follow Section 2B 3. Random Vibration, Performed in shipping box. All six of the container sides will be tested for 30 minutes/side for Single Containers. (Total 180 mins) 4. Drop - Sequence for Single Containers on 1 corner, 3 edges, 6 flats. Drop heights should follow ASTM Packaged Product Test Procedure. Drop Height based on guidelines listed in spec. Test sequence per specification. 5. Compression - Apply and hold force= $W \times (S-1) \times F$. No damage on package. W: total weight of packaged product S: total number of packaged products in a stack F: compensating factor; a minimum of 4 (all this is in the spec but listed for reference) • (3E) Palletized Package: 1. Use section 3E 2. Vibration - 60 mins on bottom side for Palletized Containers. 3. Drop – Sequence for Palletized Containers: a. Opposite edge with a wooden fixture 4 inch in height. b. 8 inches off the surface and drop for 4 edges. • Post Test Functional Requirements: 1. Hydrostatic pressure test as outlined in test 11.1.5 • No physical damage from visual inspection during chamber and post test, including but not limited to no leakage, rust or oxidation, etc.	ISTA 2A/2B/3E
11.1-17 Operating Temperature Testing	• Test Conditions: 1. Ambient: 62°C @ 15%RH: 16hrs; Fluid temp 62°C ± 2°C 2. Ambient: -5°C: 16hrs; Fluid temp -5°C ± 2°C 3. Air Ramp Rates: 10°C/hr & 20%RH/hr max. (humidity control at -5°C not required) 4. Fluid Ramp rate: 10°C/hr max ramp • Socket and plug assemblies mated at 87.4 mm mate distance • Fluid: PG25 with minimum 55 PSIG • Post Test Functional Requirements: 1. Hydrostatic pressure test as outlined in test 11.1-5 • Same supplier and interoperability testing required, a minimum of one sample of each same supplier and interop combination required	

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	Pass/Fail: No physical damage from visual inspection, including but not limited to no leakage, rust or oxidation, etc.	
11.1-18 Operating Humidity Testing	- Test Conditions: 40C @ 93%RH: 16hrs 40C @ 10%RH: 16hrs - Fluid Temp for all tests: 23 ± 10C during dwell; Fluid Ramp rate: 10C/hr max ramp - Air Ramp Rates: 10C/hr & 20%RH/hr max - Socket and plug assemblies mated at 87.4 mm mate distance - Fluid: PG25 with minimum 55 PSIG - Post Test Functional Requirements: - Hydrostatic pressure test as outlined in test 11.1-5 - Same supplier and interoperability testing required, a minimum of one sample of each same supplier and interop combination required - Pass/Fail: No physical damage from visual inspection during chamber and post test, including but not limited to no leakage, rust or oxidation, etc.	
11.1-19 Non-Operating Temperature	- Test Conditions: 72C,15%RH: 72hrs, 20C/hr & 20%RH/hr max -42C: 72hrs, 20C/hr max - Tubing attached to plug barb during chamber test and to be connected to fluid source on opposite end for post functional test - Post Test Functional Requirements: - Hydrostatic pressure test as outlined in test 11.1-5 - Pass post test listed above and no physical damage from visual inspection, including but not limited to no leakage, rust or oxidation, etc. - Same supplier only, one pair of valves required, samples for nonoperating temperature and nonoperating humidity to be the same exact parts (serial no) - PTFE tubing to be connected to plug during test and then used for post test functional testing.	
11.1-20 Non-Operating Humidity	- Test Conditions: 40C, 95% RH: 120hrs, 20% RH/hr max 40C, 5% RH: 96hrs, 20% RH/hr max - Tubing attached to plug barb during chamber test and to be connected to fluid source on opposite end for post functional test - Post Test Functional Requirements: - Hydrostatic pressure test as outlined in test 11.1-5 - Pass/Fail: Pass post test listed above and no physical damage from visual inspection, including but not limited to no leakage, rust or oxidation, etc. - Same supplier only, one pair of valves required, samples for nonoperating temperature and nonoperating humidity to be the same exact parts (serial no) - PTFE tubing to be connected to plug during test and then used for post test functional testing. Tubing should be replaced prior to this test if the fixture end of tubing needs to be disconnected	

	for this test. If it is does not need to be replaced, the same exact tubing shall be used.	
11.1-21 Salt Spray (Corrosion resistance)	• Test Conditions: 1. Follow ASTM B117 spec 2. Duration: 72 hrs 3. Position of the part under test can be placed vertically or horizontally with thread cap on. Parts should be tested loose (socket and plug assemblies un-mated.) 4. Tubing attached to plug barb during chamber test and to be connected to fluid source on opposite end for post functional test • Post Test Functional Requirements: 1. Hydrostatic pressure test as outlined in test 11.1-5 • Pass/Fail requirements: Pass post test listed above after the chamber test. Leakage is not allowed during post test. Sample continues to operate as intended as listed above. Cosmetic issues including rust or discoloration are acceptable as long as valve performance is not affected. • Same supplier only, minimum of one pair of valves required • PTFE tubing to be connected to plug during test and then used for post test functional testing	ASTM B117

12. Test Fixture Requirements

Fixture Requirements are included in the collateral with this specification. Refer to the document for details.

13. Environmental Regulations/Environmental Requirements

- RoHS Directive (2015/863/EU); aims to reduce the environmental impact of EEE by restricting the use of certain substances during manufacture.
- REACH Regulation (EC) No 1907/2006; the risk assessment for the restriction of PFAS, and registration with the European Chemicals Agency (ECHA), evaluation, authorization and restriction of chemicals.
- WEEE Directive (2012/19/EU); aims to reduce the environmental impact of EEE by restricting the use of certain substances during manufacture.
- **Substances of Concern In articles as such or in complex objects (Products):** product shall not contain any substances regulated by EPA 40 CFR751

Table 13 Environmental Specifications

Condition	Specification
Operating Temperate	-5C to 60C with fluid fluid*

Nonoperating Temperature	-40C to 70C, without fluid
Operating Humidity	10% to 93%, noncondensing, with fluid*
Nonoperating Humidity	5% to 95%, noncondensing, without fluid
Salt Spray	Per ASTM B117, without fluid

* Testing was performed with Dowfrost LC25. Usage of other fluids may affect this working range and will need to be tested by end user.

14. Safety

Test items	Requirements	Reference	Criteria
Hydrostatic pressure (Normal operation)	QC shall be designed to meet three times the maximum working pressure at normal operating condition (PSI to meet: 150)	UL62368-1	During and after tests, there shall be no rupture, no leaks and no loosening of any connection or part.
Hydrostatic pressure (Abnormal operation)	QC shall be designed to meet two times the maximum working pressure after single fault condition is applied (PSI to meet: 100)	UL62368-1	During and after tests, there shall be no rupture, no leaks and no loosening of any connection or part.
Force test	A force of $30N \pm 3N$ is applied to any accessible part for a duration of 5 sec.	UL62368-1	After tests, there shall be no rupture, no leaks and no loosening of any connection or part.

15. Prescribed Materials

15.1 Non-Wetted Materials

- ⌘ Plug cup to be zinc plated steel or stainless steel
- ⌘ Panel nut to be zinc plated steel
- ⌘ All other non-wetted materials are not defined
- ⌘ All components to be RoHs compliant

15.2 Wetted Materials

- ⌘ *Plug and socket components: SS303*
- ⌘ Springs (Internal and external) any of the following: SS302, UNS S30203, SS304, UNS S30403, SS316, UNS S31603, SS321, UNS S32103, SS410, UNS17700
- ⌘ All wetted stainless steel components to be passivated
- ⌘ Seals to be EPDM
- ⌘ All components to be RoHs compliant -

16. References

1. Open Rack V3 Base Specification
2. Meta Open Rack Frame V3 Specification
3. Open Rack V3 Blind Mate Manifold Base Specification
4. Open Rack V3 IT Gear Design Guide
5. ISO 18869:2017
6. UL62368-1 2nd edition
7. ASTM B117